

[This question paper contains 28 printed pages.]

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Your Roll No.....

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Name of the Paper : Introductory Econometrics

Name of the Course : B.A. (Hons.) Economics (NEP-UGCF-2022)-  
DSCC-12

Semester : IV

Duration : 3 Hour

Maximum Marks : 90

**Instructions for Candidates**

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Question 1 is compulsory. Answer any **four** questions from the rest of the **six** questions. In total, **five** questions to be answered.
3. Use of simple calculator is permitted.
4. All intermediate calculations should be rounded off to four decimal places. The values provided in statistical tables should not be rounded off. All final calculations should be rounded off to two decimal places.
5. Answers may be written either in English or Hindi; but the same medium should be used throughout the paper.

**छात्रों के लिए निर्देश**

1. इस प्रश्न-पत्र के मिलते ही ऊपर दिए गए निर्धारित स्थान पर अपना अनुक्रमांक लिखिए।
2. प्रश्न 1 अनिवार्य है। शेष छह प्रश्नों में से किसी चार प्रश्नों के उत्तर दीजिए। कुल पाँच प्रश्नों के उत्तर देने हैं।
3. साधारण कैलकुलेटर उपयोग की अनुमति है।
4. सभी मध्यवर्ती गणनाएँ चार दशमलव स्थानों तक पूर्णांकित की जानी चाहिए। सांख्यिकीय तालिकाओं में दिए गए मानों को पूर्णांकित नहीं किया जाना चाहिए। सभी अंतिम गणनाएँ दो दशमलव स्थानों तक पूर्णांकित की जानी चाहिए।
5. इस प्रश्न-पत्र का उत्तर अंग्रेजी या हिंदी किसी एक भाषा में दीजिए, लेकिन सभी उत्तरों का माध्यम एक ही होना चाहिए।

P.T.O.

1. State whether the following statements are true or false. Give reasons for your answer.

- (a) For an OLS estimated regression equation,  $\hat{Y}_i = b_1 + b_2X_i$ , the sum of the product of the residuals and mean deviation of the explanatory variable is zero.
- (b) Ceteris paribus, the higher the VIF (Variance Inflation Factor), the smaller is the variances of OLS estimators.
- (c) For an OLS estimated regression equation,  $\hat{Y}_i = b_1 + b_2X_i$ , if we multiply both Y and X by 1000 and re-estimate two variable regression model, the intercept coefficient will increase by 1000 times.
- (d) For a three variable regression model, TSS is always equal to the sum of ESS and RSS.
- (e) It can easily happen that F statistic is significant while t statistics of explanatory variables are not significant.
- (f) In a regression model with intercept, number of dummies for each qualitative variable must be one less than the number of categories of that variable. (3×6)

निम्नलिखित कथन सत्य हैं या असत्य। कारण सहित उत्तर दीजिए।

- (क) एक ओएलएस (OLS) के लिए अनुमानित प्रतिगमन समीकरण  $\hat{Y}_i = b_1 + b_2X_i$  है, अवशिष्ट के गुणनफल का योग और व्याख्यात्मक चर का माध्य विचलन शून्य है।

- (ख) सेटेरिस परिबस (Ceteris paribus), VIF (विचरण मुद्रास्फीति कारक) जितना उच्च होगा, ओएलएस अनुमानक के प्रसरण उतने ही छोटे होंगे।
- (ग) एक ओएलएस के लिए अनुमानित प्रतिगमन समीकरण  $\hat{Y}_i = b_1 + b_2 X_i$  है, यदि हम  $Y$  और  $X$  दोनों को 1000 से गुणा करते हैं और दो चर प्रतिगमन मॉडल का पुनः अनुमान लगाते हैं, तो अवरोधन गुणांक 1000 गुना बढ़ जाएगा।
- (घ) तीन चर प्रतिगमन मॉडल के लिए, टीएसएस हमेशा ईएसएस और आरएसएस के योग के बराबर होता है।
- (ङ) यह आसानी से हो सकता है कि  $F$  सांख्यिकी महत्वपूर्ण है जबकि व्याख्यात्मक चर की  $t$  सांख्यिकी महत्वपूर्ण नहीं है।
- (च) अवरोधन के साथ एक प्रतिगमन मॉडल में, प्रत्येक गुणात्मक चर के लिए डी की संख्या उस चर की श्रेणियों की संख्या से एक कम होनी चाहिए।
2. Using data for 110 schools the following model of monthly expenditure incurred by a school was estimated :

$$\log(\bar{Y})_i = 138.8 - 1.571 D_i + 0.808 X_i - 0.0944 D_i X_i$$

$$t = \quad (2.864) \quad (-1.06) \quad (13.466) \quad (-2.04)$$

$Y_i$  = annual expenditure incurred by a school, in thousands of rupees  $X_i$  = number

of students in the school  $D_i = \begin{cases} 1 & \text{if government school} \\ 0 & \text{if private school} \end{cases}$

- (i) Write the regression equation for government school and private school separately. Also, interpret the slope coefficient of two categories.

- (ii) Check the statistical significance of differential intercept coefficient and slope drifter at 10% level of significance. Based on the conclusion of these tests, draw the population regression lines for the government and private schools.
- (iii) Test the hypothesis that the population error term is normally distributed at 1% level of significance when it is given that for the residuals, skewness is 0.28 and the value of kurtosis is 2. Also, in classical linear regression model, what is the rationale for the assumption that population error term follows normal distribution? (6,6,6)

110 स्कूलों के डेटा का उपयोग करते हुए, किसी स्कूल द्वारा किए गए मासिक व्यय के निम्नलिखित मॉडल का अनुमान लगाया गया था :

$$\log(Y)_i = 138.8 - 1.571 D_i + 0.808 X_i - 0.0944 D_i X_i$$

$$t = \quad (2.864) \quad (-1.06) \quad (13.466) \quad (-2.04)$$

$Y_i$  = स्कूल द्वारा किया गया वार्षिक व्यय, हजार रुपये में

$X_i$  = स्कूल में विद्यार्थियों की संख्या

$$D_i = \begin{cases} 1 & \text{यदि सरकारी स्कूल है} \\ 0 & \text{यदि निजी स्कूल है} \end{cases}$$

- (i) सरकारी स्कूल और निजी स्कूल के लिए अलग-अलग प्रतिगमन समीकरण लिखिए।

इसके अलावा, दोनों श्रेणियों के ढलान गुणांक की व्याख्या कीजिए।

(ii) महत्व के 10% स्तर पर अंतर अवरोधन गुणांक और ढलान डिफ्टर के सांख्यिकीय महत्व की जाँच कीजिए। इन परीक्षणों के निष्कर्ष के आधार पर, सरकारी और निजी स्कूलों के लिए जनसंख्या प्रतिगमन रेखाएँ खींचिए।

(iii) इस परिकल्पना का परीक्षण कीजिए कि जनसंख्या त्रुटि शब्द सामान्य रूप से महत्व के 1% स्तर पर वितरित किया जाता है जब अवशिष्टों, विषमता के लिए 0.28 है और पृथुशीर्षत्व का मान 2 दिया जाता है। इसके अलावा, शास्त्रीय रैखिक प्रतिगमन मॉडल में, इस धारणा के लिए तर्क क्या है कि जनसंख्या त्रुटि शब्द सामान्य वितरण का अनुसरण करता है?

3. (a) For estimating the Phillips curves for the United States from 1958 to 1969 the following regression was obtained :

$$\begin{aligned} \hat{Y}_t &= -0.2594 + 20.5880 \frac{1}{X_t} \\ t &= (-0.2572) \quad (4.3996) \\ R^2 &= 0.6594, d = 0.6394 \end{aligned}$$

Y = Rate of change of money wages in percent

X = Unemployment rate in percent

- (i) Interpret the intercept estimate. Is there any evidence of first order autocorrelation in the residuals at 5% level of significance?
  - (ii) Outline the method of estimation that will produce BLUE estimators in the presence of first order autocorrelation.
- (b) Consider the following regression function on sales revenue for a particular firm for last 10 months, as estimated by OLS :

$$\hat{Y}_t = 51.508 + 0.054 X_t$$

$$se = (4.03) \quad (0.04)$$

$$\widehat{\sigma_u^2} = 1.02$$

$$\bar{X} = 25 \quad \sum X_t^2 = 6500$$

Y is the monthly sales revenue in billions of dollars

X is the monthly expenditure on advertising in millions of dollars.

Find the predicted mean sales revenue if the advertising expenditure for the firm in the next month is 30 million dollars. Also, find the 99% confidence interval for the true predicted mean of sales revenue corresponding to 30 million dollars advertising expenditure. (6+6,6)

(क) 1958 से 1969 तक संयुक्त राज्य अमेरिका के लिए फिलिप्स वक्र का अनुमान लगाने के लिए निम्नलिखित प्रतिगमन प्राप्त किया गया था :

$$\hat{Y}_t = -0.2594 + 20.5880 \frac{1}{X_t}$$

$$t = (-0.2572) \quad (4.3996)$$

$$R^2 = 0.6594, d = 0.6394$$

Y = मुद्रा मजदूरी के परिवर्तन की दर प्रतिशत में

X = बेरोजगारी दर प्रतिशत में

(i) अवरोधन अनुमान की व्याख्या कीजिए। क्या 5% स्तर के महत्व पर अवशिष्टों में प्रथम क्रम के स्वतः सहसंबंध का कोई प्रमाण है?

(ii) पहले क्रम के स्वसहसंबंध की उपस्थिति में 'ब्लू' अनुमानक देने वाली अनुमान विधि की रूपरेखा तैयार कीजिए।

(ख) पिछले 10 महीनों के लिए किसी विशेष फर्म के लिए बिक्री राजस्व पर निम्नलिखित प्रतिगमन फलन पर विचार कीजिए, जैसा कि ओएलएस द्वारा अनुमानित है :

$$\hat{Y}_t = 51.508 + 0.054 X_t$$

$$se = (4.03) \quad (0.04)$$

$$\widehat{\sigma_u^2} = 1.02$$

$$\bar{X}=25 \quad \sum X_t^2 = 6500$$

Y बिलियन डॉलर में मासिक बिक्री राजस्व है

X मिलियन डॉलर में विज्ञापन पर मासिक खर्च है

यदि अगले महीने में फर्म का विज्ञापन व्यय 30 मिलियन डॉलर है, तो अनुमानित औसत बिक्री राजस्व ज्ञात कीजिए। इसके अलावा, 30 मिलियन डॉलर के विज्ञापन व्यय के अनुरूप बिक्री राजस्व के वास्तविक अनुमानित औसत के लिए 99% कॉन्फिडेंस इंटरवल ज्ञात कीजिए।

4. In a regression of housing expenditure in rupees ( $Y_i$ ) on annual income of families in rupees ( $X_i$ ) for a sample of 27 families, the following results were obtained :

| Variable      | Coefficients | Standard Error |
|---------------|--------------|----------------|
| X             | 0.121        | 0.009          |
| Constant      | 3.803        | 4.570          |
| n = 27        |              |                |
| $R^2 = 0.776$ |              |                |

On plotting the residuals against  $X_i$ , it was found that the variance of the residuals increased with  $X_i$ .

- (i) What problem does this indicate? Name any one test for its detection and explain the steps to conduct this test.
- (ii) What are the consequences of this problem for OLS estimators? Which type of dataset is more likely to be characterized by this problem?
- (iii) Explain the estimation process of Weighted Least Squares with unknown error variances in this context where error variance is proportional to  $X^2$ . Also, mention which is the intercept and slope parameter in the transformed model and how do we get back to the original model. (6,6,6)

27 परिवारों के नमूने के लिए ( $X_i$ ) रुपये की परिवारों की वार्षिक आय में से ( $Y_i$ ) रुपये के आवास व्यय के प्रतिगमन में, निम्नलिखित परिणाम प्राप्त हुए थे :

| चर            | गुणांक | मानक त्रुटि |
|---------------|--------|-------------|
| X             | 0.121  | 0.009       |
| स्थिरांक      | 3.803  | 4.570       |
| n = 27        |        |             |
| $R^2 = 0.776$ |        |             |

$X_1$  के समक्ष अवशिष्टों को रखे जाने पर यह पाया गया कि अवशिष्टों के विचरण में  $X_1$  की वृद्धि हुई है।

- (i) यह किस समस्या को इंगित करता है? इसकी पहचान के लिए किसी एक परीक्षण का नाम बताइए और इस परीक्षण को करने के चरणों की व्याख्या कीजिए।
  - (ii) ओएलएस अनुमानक के लिए इस समस्या के परिणाम क्या हैं? इस समस्या से किस प्रकार के आंकड़ा समुच्चय की विशेषता होने की अधिक संभावना है?
  - (iii) इस संदर्भ में अज्ञात त्रुटि भिन्नताओं के साथ भारित कम से कम वर्गों की अनुमान प्रक्रिया की व्याख्या कीजिए जहां त्रुटि विचरण  $X^2$  के समानुपाती है। इसके अलावा, रूपांतरित मॉडल में अवरोधन और ढलान पैरामीटर कौन-सा होता है और हम मूल मॉडल पर वापस कैसे आते हैं, इसका उल्लेख कीजिए।
5. (a) Suppose a third-degree polynomial regression was fitted to a cost-output model for 18 firms and the following results were obtained :

$$\widehat{Cost}_i = 141.7667 + 0.208 Q_i - 0.086 Q_i^2 + 0.0054 Q_i^3$$

$$se = (6.37) \quad (4.78) \quad (2.98) \quad (6.98)$$

$Q_i$  is the output of the  $i^{th}$  firm.

- (i) If cost curves are to have U-shaped average and marginal cost curves, then what are your a priori expectations about the intercept and slope estimators? Check if these a priori expectations are satisfied for the given model.
- (ii) Interpret the estimated slope coefficient of  $Q^2$  and test its significance at 10% level of significance.
- (b) Let  $Y$  be the Gross National Product,  $X_2$  be the Exports,  $X_3$  be the Imports and  $X_4$  be the Net Exports in the following relationship:

$$Y_i = B_1 + B_2 X_{2i} + B_3 X_{3i} + B_4 X_{4i} + U_i$$

Which assumptions of Classical Linear Regression Model is violated here? If you estimate this equation by ordinary least squares, then which of the parameters can be estimated? Explain. (6+6,6)

- (क) मान लीजिए कि एक तृतीय-डिग्री बहुपद प्रतिगमन को 18 फर्मों के लिए लागत-आउटपुट मॉडल में फिट किया गया था जिससे निम्नलिखित परिणाम प्राप्त किए गए थे :

$$\widehat{Cost}_i = 141.7667 + 0.208 Q_i - 0.086 Q_i^2 + 0.0054 Q_i^3$$

$$se = (6.37) \quad (4.78) \quad (2.98) \quad (6.98)$$

$Q_i$ ,  $i^{th}$  फर्म का आउटपुट है।

- (i) यदि लागत वक्रों में यू-आकार का औसत और सीमांत लागत वक्र होते हैं, तो आप अवरोधन और ढलान अनुमानक के बारे में प्राथमिकता की क्या अपेक्षा कर रहे हैं? क्या दी गई प्राथमिकता अपेक्षाएं दिए गए मॉडल को संतुष्ट करती हैं।
- (ii)  $Q^2$  के अनुमानित ढलान गुणांक की व्याख्या कीजिए और महत्ता के 10% स्तर पर इसके महत्ता का परीक्षण कीजिए।
- (ख) मान लीजिए  $Y$  सकल राष्ट्रीय उत्पाद है,  $X_2$  निर्यात है, आयात  $X_3$  है और निम्नलिखित संबंध में शुद्ध निर्यात  $X_4$  हैं:

$$Y_i = B_1 + B_2 X_{2i} + B_3 X_{3i} + B_4 X_{4i} + U_i$$

यहां शास्त्रीय रेखीय प्रतिगमन मॉडल की कौन-सी मान्यताओं का उल्लंघन किया गया है? यदि आप इस समीकरण का अनुमान साधारण न्यूनतम वर्गों से लगाते हैं, तो किस पैरामीटर का अनुमान लगाया जा सकता है? स्पष्ट कीजिए।

6. (a) For a particular cafeteria, the following equation was estimated using yearly data on cups of chocolate flavoured cold coffee sold :

$$\log(\bar{Q}_t) = 1.534 - 0.250 \log(P_t) + 0.025 \log(P_t^{**})$$

$$se = (0.2001) \quad (0.240) \quad (0.016)$$

$$R^2 = 0.804 \quad TSS = 4803$$

Where

$Q_t$  = cups of chocolate flavoured cold coffee sold

$P_t$  = price (in rupees) per cup of chocolate flavoured cold coffee

$P_t^{**}$  = price (in rupees) per cup of vanilla flavoured cold coffee

$t$  = 1991-2023

Note that all cups in the cafeteria are of same size.

- (i) What can you say about the own-price and cross-price elasticity of demand for chocolate flavoured coffee? Test the hypothesis that the own price elasticity of demand of chocolate flavoured coffee is unitary elastic at 10% level of significance.
- (ii) Draw the ANOVA table for the regression equation. Also calculate  $\overline{R}^2$ .
- (b) Explain the statement "It is better to include irrelevant variables than to exclude relevant ones". (6+6,6)

(क) एक विशेष कैफेटेरिया के लिए, चॉकलेट फ्लेवर्ड कोल्ड कॉफी के बेचे गए कप से संबंधित वार्षिक डेटा का उपयोग करके निम्नलिखित समीकरण का अनुमान लगाया गया था :

$$\log(\widehat{Q}_t) = 1.534 - 0.250 \log(P_t) + 0.025 \log(P_t^{**})$$

$$se = (0.2001) \quad (0.240) \quad (0.016)$$

$$R^2 = 0.804 \quad TSS = 4803$$

जहाँ

$Q_t$  = चॉकलेट फ्लेवर्ड कोल्ड कॉफी के बेचे गए कप

$P_t$  = प्रति कप चॉकलेट फ्लेवर्ड कोल्ड कॉफी की कीमत (रुपये में)

$P_t^{**}$  = वनीला फ्लेवर्ड कोल्ड कॉफी की प्रति कप कीमत (रुपये में)

$t$  = 1991-2023

नोट कीजिए कि कैफेटेरिया में सभी कप एक ही आकार के हैं।

- (i) चॉकलेट फ्लेवर्ड कॉफी की माँग की स्वयं की कीमत और क्रॉस-कीमत लोच के बारे में आप क्या कहेंगे? इस परिकल्पना का परीक्षण कीजिए कि चॉकलेट फ्लेवर्ड कॉफी की माँग की अपनी कीमत लोच महत्ता के 10% स्तर पर एकात्मक लोचदार है।
- (ii) प्रतिगमन समीकरण के लिए एनोवा तालिका तैयार कीजिए साथ ही  $\bar{R}^2$  की भी गणना कीजिए।
- (ख) “प्रासंगिक चर को बाहर करने की तुलना में अप्रासंगिक चर को शामिल करना बेहतर है”। इस कथन की व्याख्या कीजिए।

7. (a) Explain the steps involved in performing Ramsey's RESET test for model specification.
- (b) Using data collected from 1990 to 2022, for a particular country, the following equation was estimated using OLS :

$$\begin{aligned} \hat{Y}_i &= 4.47 + 0.34 \log X_{2i} - 1.22 \log X_{3i} \\ t &= (4.28) \quad (5.31) \quad (-0.98) \\ R^2 &= 0.9987 \end{aligned}$$

Y = Child Mortality Rate

$X_2$  = Female Literacy Rate

$X_3$  = Per Capita GNP

- (i) Are the regression results in conformity with your a priori expectations? If not, what can be the reason behind it? Explain any three remedies to overcome this problem.

P.T.O.

- (ii) Now suppose the researcher regressed female literacy rate on per capita GNP and in this second regression, the  $R^2$  is 0.5284. Do you think that the reason that you suggested in (i) is significantly present in the model, at 5% level of significance? If your answer is yes, then do you think that the presence of this is necessarily bad? (6,6+6)

- (क) मॉडल विनिर्देश के लिए रैमसे के रीसेट परीक्षण करने में शामिल चरणों की व्याख्या कीजिए।  
 (ख) किसी विशेष देश के लिए 1990 से 2022 तक एकत्र किए गए डेटा का उपयोग करते हुए, ओएलएस का उपयोग करके निम्नलिखित समीकरण का

$$\hat{Y}_i = 4.47 + 0.34 \log X_{2i} - 1.24 \log X_{3i}$$

$$t = (4.28) \quad (5.31) \quad (-0.98)$$

$$R^2 = 0.9987$$

$Y$  = बाल मृत्यु दर

$X_2$  = महिला साक्षरता दर

$X_3$  = प्रति व्यक्ति सकल राष्ट्रीय उत्पाद (जीएनपी)

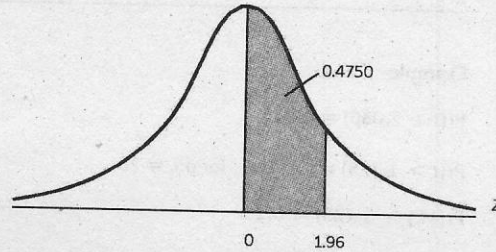
- (i) क्या प्रतिगमन परिणाम आपकी प्राथमिकता अपेक्षाओं के अनुरूप हैं? यदि नहीं, तो इसके पीछे क्या कारण हो सकता है? इस समस्या को दूर करने के कोई तीन उपाय बताइए।
- (ii) अब मान लीजिए कि शोधकर्ता ने प्रति व्यक्ति जीएनपी पर महिला साक्षरता दर को छोड़ दिया और इस दूसरे प्रतिगमन में,  $R^2$ , 0.5284 है। क्या आपको लगता है कि आपने (i) में जो कारण सुझाया है, वह मॉडल में 5% महत्ता स्तर पर महत्वपूर्ण रूप से मौजूद है? यदि आपका उत्तर हाँ है, तो क्या आपको लगता है कि इसकी उपस्थिति निश्चित रूप से खराब है?

TABLE E-1a AREAS UNDER THE STANDARDIZED NORMAL DISTRIBUTION

## Example

$$\Pr(0 \leq Z \leq 1.96) = 0.4750$$

$$\Pr(Z \geq 1.96) = 0.5 - 0.4750 = 0.025$$



| Z   | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.0 | .0000 | .0040 | .0080 | .0120 | .0160 | .0199 | .0239 | .0279 | .0319 | .0359 |
| 0.1 | .0398 | .0438 | .0478 | .0517 | .0557 | .0596 | .0636 | .0675 | .0714 | .0753 |
| 0.2 | .0793 | .0832 | .0871 | .0910 | .0948 | .0987 | .1026 | .1064 | .1103 | .1141 |
| 0.3 | .1179 | .1217 | .1255 | .1293 | .1331 | .1368 | .1406 | .1443 | .1480 | .1517 |
| 0.4 | .1554 | .1591 | .1628 | .1664 | .1700 | .1736 | .1772 | .1808 | .1844 | .1879 |
| 0.5 | .1915 | .1950 | .1985 | .2019 | .2054 | .2088 | .2123 | .2157 | .2190 | .2224 |
| 0.6 | .2257 | .2291 | .2324 | .2357 | .2389 | .2422 | .2454 | .2486 | .2517 | .2549 |
| 0.7 | .2580 | .2611 | .2642 | .2673 | .2704 | .2734 | .2764 | .2794 | .2823 | .2852 |
| 0.8 | .2881 | .2910 | .2939 | .2967 | .2995 | .3023 | .3051 | .3078 | .3106 | .3133 |
| 0.9 | .3159 | .3186 | .3212 | .3238 | .3264 | .3289 | .3315 | .3340 | .3365 | .3389 |
| 1.0 | .3413 | .3438 | .3461 | .3485 | .3508 | .3531 | .3554 | .3577 | .3599 | .3621 |
| 1.1 | .3643 | .3665 | .3686 | .3708 | .3729 | .3749 | .3770 | .3790 | .3810 | .3830 |
| 1.2 | .3849 | .3869 | .3888 | .3907 | .3925 | .3944 | .3962 | .3980 | .3997 | .4015 |
| 1.3 | .4032 | .4049 | .4066 | .4082 | .4099 | .4115 | .4131 | .4147 | .4162 | .4177 |
| 1.4 | .4192 | .4207 | .4222 | .4236 | .4251 | .4265 | .4279 | .4292 | .4306 | .4319 |
| 1.5 | .4332 | .4345 | .4357 | .4370 | .4382 | .4394 | .4406 | .4418 | .4429 | .4441 |
| 1.6 | .4452 | .4463 | .4474 | .4484 | .4495 | .4505 | .4515 | .4525 | .4535 | .4545 |
| 1.7 | .4454 | .4564 | .4573 | .4582 | .4591 | .4599 | .4608 | .4616 | .4625 | .4633 |
| 1.8 | .4641 | .4649 | .4656 | .4664 | .4671 | .4678 | .4686 | .4693 | .4699 | .4706 |
| 1.9 | .4713 | .4719 | .4726 | .4732 | .4738 | .4744 | .4750 | .4756 | .4761 | .4767 |
| 2.0 | .4772 | .4778 | .4783 | .4788 | .4793 | .4798 | .4803 | .4808 | .4812 | .4817 |
| 2.1 | .4821 | .4826 | .4830 | .4834 | .4838 | .4842 | .4846 | .4850 | .4854 | .4857 |
| 2.2 | .4861 | .4864 | .4868 | .4871 | .4875 | .4878 | .4881 | .4884 | .4887 | .4890 |
| 2.3 | .4893 | .4896 | .4898 | .4901 | .4904 | .4906 | .4909 | .4911 | .4913 | .4916 |
| 2.4 | .4918 | .4920 | .4922 | .4925 | .4927 | .4929 | .4931 | .4932 | .4934 | .4936 |
| 2.5 | .4938 | .4940 | .4941 | .4943 | .4945 | .4946 | .4948 | .4949 | .4951 | .4952 |
| 2.6 | .4953 | .4955 | .4956 | .4957 | .4959 | .4960 | .4961 | .4962 | .4963 | .4964 |
| 2.7 | .4965 | .4966 | .4967 | .4968 | .4969 | .4970 | .4971 | .4972 | .4973 | .4974 |
| 2.8 | .4974 | .4975 | .4976 | .4977 | .4977 | .4978 | .4979 | .4979 | .4980 | .4981 |
| 2.9 | .4981 | .4982 | .4982 | .4983 | .4984 | .4984 | .4985 | .4985 | .4986 | .4986 |
| 3.0 | .4987 | .4987 | .4987 | .4988 | .4988 | .4989 | .4989 | .4989 | .4990 | .4990 |

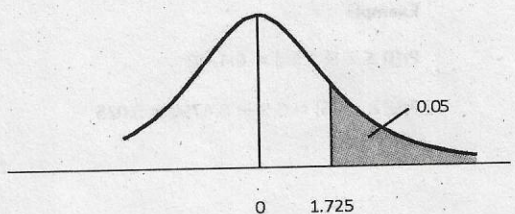
Note: This table gives the area in the right-hand tail of the distribution (i.e.,  $Z \geq 0$ ). But since the normal distribution is symmetrical about  $Z = 0$ , the area in the left-hand tail is the same as the area in the corresponding right-hand tail. For example,  $P(-1.96 \leq Z \leq 0) = 0.4750$ . Therefore,  $P(-1.96 \leq Z \leq 1.96) = 2(0.4750) = 0.95$ .

TABLE E-2 PERCENTAGE POINTS OF THE  $t$  DISTRIBUTION**Example**

$$\Pr(t > 2.086) = 0.025$$

$$\Pr(t > 1.725) = 0.05 \quad \text{for d.f.} = 20$$

$$\Pr(|t| > 1.725) = 0.10$$

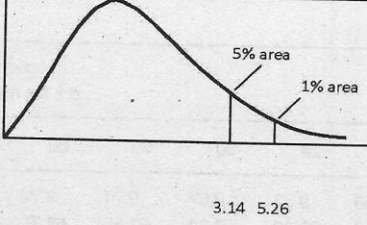


| Pr<br>d.f. | 0.25<br>0.50 | 0.10<br>0.20 | 0.05<br>0.10 | 0.025<br>0.05 | 0.01<br>0.02 | 0.005<br>0.010 | 0.001<br>0.002 |
|------------|--------------|--------------|--------------|---------------|--------------|----------------|----------------|
| 1          | 1.000        | 3.078        | 6.314        | 12.706        | 31.821       | 63.657         | 318.31         |
| 2          | 0.816        | 1.886        | 2.920        | 4.303         | 6.965        | 9.925          | 22.327         |
| 3          | 0.765        | 1.638        | 2.353        | 3.182         | 4.541        | 5.841          | 10.214         |
| 4          | 0.741        | 1.533        | 2.132        | 2.776         | 3.747        | 4.604          | 7.173          |
| 5          | 0.727        | 1.476        | 2.015        | 2.571         | 3.365        | 4.032          | 5.893          |
| 6          | 0.718        | 1.440        | 1.943        | 2.447         | 3.143        | 3.707          | 5.208          |
| 7          | 0.711        | 1.415        | 1.895        | 2.365         | 2.998        | 3.499          | 4.785          |
| 8          | 0.706        | 1.397        | 1.860        | 2.306         | 2.896        | 3.355          | 4.501          |
| 9          | 0.703        | 1.383        | 1.833        | 2.262         | 2.821        | 3.250          | 4.297          |
| 10         | 0.700        | 1.372        | 1.812        | 2.228         | 2.764        | 3.169          | 4.144          |
| 11         | 0.697        | 1.363        | 1.796        | 2.201         | 2.718        | 3.106          | 4.025          |
| 12         | 0.695        | 1.356        | 1.782        | 2.179         | 2.681        | 3.055          | 3.930          |
| 13         | 0.694        | 1.350        | 1.771        | 2.160         | 2.650        | 3.012          | 3.852          |
| 14         | 0.692        | 1.345        | 1.761        | 2.145         | 2.624        | 2.977          | 3.787          |
| 15         | 0.691        | 1.341        | 1.753        | 2.131         | 2.602        | 2.947          | 3.733          |
| 16         | 0.690        | 1.337        | 1.746        | 2.120         | 2.583        | 2.921          | 3.686          |
| 17         | 0.689        | 1.333        | 1.740        | 2.110         | 2.567        | 2.898          | 3.646          |
| 18         | 0.688        | 1.330        | 1.734        | 2.101         | 2.552        | 2.878          | 3.610          |
| 19         | 0.688        | 1.328        | 1.729        | 2.093         | 2.539        | 2.861          | 3.579          |
| 20         | 0.687        | 1.325        | 1.725        | 2.086         | 2.528        | 2.845          | 3.552          |
| 21         | 0.686        | 1.323        | 1.721        | 2.080         | 2.518        | 2.831          | 3.527          |
| 22         | 0.686        | 1.321        | 1.717        | 2.074         | 2.508        | 2.819          | 3.505          |
| 23         | 0.685        | 1.319        | 1.714        | 2.069         | 2.500        | 2.807          | 3.485          |
| 24         | 0.685        | 1.318        | 1.711        | 2.064         | 2.492        | 2.797          | 3.467          |
| 25         | 0.684        | 1.316        | 1.708        | 2.060         | 2.485        | 2.787          | 3.450          |
| 26         | 0.684        | 1.315        | 1.706        | 2.056         | 2.479        | 2.779          | 3.435          |
| 27         | 0.684        | 1.314        | 1.703        | 2.052         | 2.473        | 2.771          | 3.421          |
| 28         | 0.683        | 1.313        | 1.701        | 2.048         | 2.467        | 2.763          | 3.408          |
| 29         | 0.683        | 1.311        | 1.699        | 2.045         | 2.462        | 2.756          | 3.396          |
| 30         | 0.683        | 1.310        | 1.697        | 2.042         | 2.457        | 2.750          | 3.385          |
| 40         | 0.681        | 1.303        | 1.684        | 2.021         | 2.423        | 2.704          | 3.307          |
| 60         | 0.679        | 1.296        | 1.671        | 2.000         | 2.390        | 2.660          | 3.232          |
| 120        | 0.677        | 1.289        | 1.658        | 1.980         | 2.358        | 2.617          | 3.160          |
| q          | 0.674        | 1.282        | 1.645        | 1.960         | 2.326        | 2.576          | 3.090          |

Note: The smaller probability shown at the head of each column is the area in one tail; the larger probability is the area in both tails.

Source: From E. S. Pearson and H. O. Hartley, eds., *Biometrika Tables for Statisticians*, vol. 1, 3rd ed., Table 12, Cambridge University Press, New York, 1966. Reproduced by permission of the editors and trustees of *Biometrika*.

TABLE E-3 UPPER PERCENTAGE POINTS OF THE F DISTRIBUTION

| Example                                                                            |                          |        |        |        |        |        |        |        |        |        |        |        |        |
|------------------------------------------------------------------------------------|--------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $\Pr(F > 1.59) = 0.25$                                                             |                          |        |        |        |        |        |        |        |        |        |        |        |        |
| $\Pr(F > 2.42) = 0.10$                                                             |                          |        |        |        |        |        |        |        |        |        |        |        |        |
| $\Pr(F > 3.14) = 0.05$                                                             |                          |        |        |        |        |        |        |        |        |        |        |        |        |
| $\Pr(F > 5.26) = 0.01$                                                             |                          |        |        |        |        |        |        |        |        |        |        |        |        |
| for d.f. $N_1 = 10$                                                                |                          |        |        |        |        |        |        |        |        |        |        |        |        |
| and $N_2 = 9$                                                                      |                          |        |        |        |        |        |        |        |        |        |        |        |        |
|  |                          |        |        |        |        |        |        |        |        |        |        |        |        |
| d.f. for<br>denom-<br>inator<br>$N_2$                                              | d.f. for numerator $N_1$ |        |        |        |        |        |        |        |        |        |        |        |        |
|                                                                                    | Pr                       | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     | 12     |
| 1                                                                                  | .25                      | 5.83   | 7.50   | 8.20   | 8.58   | 8.82   | 8.98   | 9.10   | 9.19   | 9.26   | 9.32   | 9.36   | 9.41   |
|                                                                                    | .10                      | 39.90  | 49.50  | 53.60  | 55.80  | 57.20  | 58.20  | 58.90  | 59.40  | 59.90  | 60.20  | 60.50  | 60.70  |
|                                                                                    | .05                      | 161.00 | 200.00 | 216.00 | 225.00 | 230.00 | 234.00 | 237.00 | 239.00 | 241.00 | 242.00 | 243.00 | 244.00 |
|                                                                                    | .01                      | 98.50  | 99.00  | 99.20  | 99.20  | 99.30  | 99.30  | 99.40  | 99.40  | 99.40  | 99.40  | 99.40  | 99.40  |
| 2                                                                                  | .25                      | 2.57   | 3.00   | 3.15   | 3.23   | 3.28   | 3.31   | 3.34   | 3.35   | 3.37   | 3.38   | 3.39   | 3.39   |
|                                                                                    | .10                      | 8.53   | 9.00   | 9.16   | 9.24   | 9.29   | 9.33   | 9.35   | 9.37   | 9.38   | 9.39   | 9.40   | 9.41   |
|                                                                                    | .05                      | 18.50  | 19.00  | 19.20  | 19.20  | 19.30  | 19.30  | 19.40  | 19.40  | 19.40  | 19.40  | 19.40  | 19.40  |
|                                                                                    | .01                      | 98.50  | 99.00  | 99.20  | 99.20  | 99.30  | 99.30  | 99.40  | 99.40  | 99.40  | 99.40  | 99.40  | 99.40  |
| 3                                                                                  | .25                      | 2.02   | 2.28   | 2.36   | 2.39   | 2.41   | 2.42   | 2.43   | 2.44   | 2.44   | 2.44   | 2.45   | 2.45   |
|                                                                                    | .10                      | 5.54   | 5.46   | 5.39   | 5.34   | 5.31   | 5.28   | 5.27   | 5.25   | 5.24   | 5.23   | 5.22   | 5.22   |
|                                                                                    | .05                      | 10.10  | 9.55   | 9.28   | 9.12   | 9.01   | 8.94   | 8.89   | 8.85   | 8.81   | 8.79   | 8.76   | 8.74   |
|                                                                                    | .01                      | 34.10  | 30.80  | 29.50  | 28.70  | 28.20  | 27.90  | 27.70  | 27.50  | 27.30  | 27.20  | 27.10  | 27.10  |
| 4                                                                                  | .25                      | 1.81   | 2.00   | 2.05   | 2.06   | 2.07   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   |
|                                                                                    | .10                      | 4.54   | 4.32   | 4.19   | 4.11   | 4.05   | 4.01   | 3.98   | 3.95   | 3.94   | 3.92   | 3.91   | 3.90   |
|                                                                                    | .05                      | 7.71   | 6.94   | 6.59   | 6.39   | 6.26   | 6.16   | 6.09   | 6.04   | 6.00   | 5.96   | 5.94   | 5.91   |
|                                                                                    | .01                      | 21.20  | 18.00  | 16.70  | 16.00  | 15.50  | 15.20  | 15.00  | 14.80  | 14.70  | 14.50  | 14.40  | 14.40  |
| 5                                                                                  | .25                      | 1.69   | 1.85   | 1.88   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   | 1.89   |
|                                                                                    | .10                      | 4.06   | 3.78   | 3.62   | 3.52   | 3.45   | 3.40   | 3.37   | 3.34   | 3.32   | 3.30   | 3.28   | 3.27   |
|                                                                                    | .05                      | 6.61   | 5.79   | 5.41   | 5.19   | 5.05   | 4.95   | 4.88   | 4.82   | 4.77   | 4.74   | 4.71   | 4.68   |
|                                                                                    | .01                      | 16.30  | 13.30  | 12.10  | 11.40  | 11.00  | 10.70  | 10.50  | 10.30  | 10.20  | 10.10  | 9.96   | 9.89   |
| 6                                                                                  | .25                      | 1.62   | 1.76   | 1.78   | 1.79   | 1.79   | 1.78   | 1.78   | 1.78   | 1.77   | 1.77   | 1.77   | 1.77   |
|                                                                                    | .10                      | 3.78   | 3.46   | 3.29   | 3.18   | 3.11   | 3.05   | 3.01   | 2.98   | 2.96   | 2.94   | 2.92   | 2.90   |
|                                                                                    | .05                      | 5.99   | 5.14   | 4.76   | 4.53   | 4.39   | 4.28   | 4.21   | 4.15   | 4.10   | 4.06   | 4.03   | 4.00   |
|                                                                                    | .01                      | 13.70  | 10.90  | 9.78   | 9.15   | 8.75   | 8.47   | 8.26   | 8.10   | 7.98   | 7.87   | 7.79   | 7.72   |
| 7                                                                                  | .25                      | 1.57   | 1.70   | 1.72   | 1.72   | 1.71   | 1.71   | 1.70   | 1.70   | 1.69   | 1.69   | 1.69   | 1.68   |
|                                                                                    | .10                      | 3.59   | 3.26   | 3.07   | 2.96   | 2.88   | 2.83   | 2.78   | 2.75   | 2.72   | 2.70   | 2.68   | 2.67   |
|                                                                                    | .05                      | 5.59   | 4.74   | 4.35   | 4.12   | 3.97   | 3.87   | 3.79   | 3.73   | 3.68   | 3.64   | 3.60   | 3.57   |
|                                                                                    | .01                      | 12.20  | 9.55   | 8.45   | 7.85   | 7.46   | 7.19   | 6.99   | 6.84   | 6.72   | 6.62   | 6.54   | 6.47   |
| 8                                                                                  | .25                      | 1.54   | 1.66   | 1.67   | 1.66   | 1.66   | 1.65   | 1.64   | 1.64   | 1.63   | 1.63   | 1.63   | 1.62   |
|                                                                                    | .10                      | 3.46   | 3.11   | 2.92   | 2.81   | 2.73   | 2.67   | 2.62   | 2.59   | 2.56   | 2.54   | 2.52   | 2.50   |
|                                                                                    | .05                      | 5.32   | 4.46   | 4.07   | 3.84   | 3.69   | 3.58   | 3.50   | 3.44   | 3.39   | 3.35   | 3.31   | 3.28   |
|                                                                                    | .01                      | 11.30  | 8.65   | 7.59   | 7.01   | 6.63   | 6.37   | 6.18   | 6.03   | 5.91   | 5.81   | 5.73   | 5.67   |
| 9                                                                                  | .25                      | 1.51   | 1.62   | 1.63   | 1.63   | 1.62   | 1.61   | 1.60   | 1.60   | 1.59   | 1.59   | 1.58   | 1.58   |
|                                                                                    | .10                      | 3.36   | 3.01   | 2.81   | 2.69   | 2.61   | 2.55   | 2.51   | 2.47   | 2.44   | 2.42   | 2.40   | 2.38   |
|                                                                                    | .05                      | 5.12   | 4.26   | 3.86   | 3.63   | 3.48   | 3.37   | 3.29   | 3.23   | 3.18   | 3.14   | 3.10   | 3.07   |
|                                                                                    | .01                      | 10.60  | 8.02   | 6.99   | 6.42   | 6.06   | 5.80   | 5.61   | 5.47   | 5.35   | 5.26   | 5.18   | 5.11   |

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| d.f. for numerator $N_1$ |        |        |        |        |        |        |        |        |        |        |        |     | d.f. for denominator $N_2$ |
|--------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-----|----------------------------|
| 15                       | 20     | 24     | 30     | 40     | 50     | 60     | 100    | 120    | 200    | 500    | $q$    | Pr  |                            |
| 9.49                     | 9.58   | 9.63   | 9.67   | 9.71   | 9.74   | 9.76   | 9.78   | 9.80   | 9.82   | 9.84   | 9.85   | .25 | 1                          |
| 61.20                    | 61.70  | 62.00  | 62.30  | 62.50  | 62.70  | 62.80  | 63.00  | 63.10  | 63.20  | 63.30  | 63.30  | .10 |                            |
| 246.00                   | 248.00 | 249.00 | 250.00 | 251.00 | 252.00 | 252.00 | 253.00 | 253.00 | 254.00 | 254.00 | 254.00 | .05 |                            |
| 3.41                     | 3.43   | 3.43   | 3.44   | 3.45   | 3.45   | 3.46   | 3.47   | 3.47   | 3.48   | 3.48   | 3.48   | .25 | 2                          |
| 9.42                     | 9.44   | 9.45   | 9.46   | 9.47   | 9.47   | 9.47   | 9.48   | 9.48   | 9.49   | 9.49   | 9.49   | .10 |                            |
| 19.40                    | 19.40  | 19.50  | 19.50  | 19.50  | 19.50  | 19.50  | 19.50  | 19.50  | 19.50  | 19.50  | 19.50  | .05 |                            |
| 99.40                    | 99.40  | 99.50  | 99.50  | 99.50  | 99.50  | 99.50  | 99.50  | 99.50  | 99.50  | 99.50  | 99.50  | .01 | 3                          |
| 2.46                     | 2.46   | 2.46   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47   | 2.47   | .25 |                            |
| 5.20                     | 5.18   | 5.18   | 5.17   | 5.16   | 5.15   | 5.15   | 5.14   | 5.14   | 5.14   | 5.14   | 5.13   | .10 |                            |
| 8.70                     | 8.66   | 8.64   | 8.62   | 8.59   | 8.58   | 8.57   | 8.55   | 8.55   | 8.54   | 8.53   | 8.53   | .05 | 4                          |
| 26.90                    | 26.70  | 26.60  | 26.50  | 26.40  | 26.40  | 26.30  | 26.20  | 26.20  | 26.20  | 26.10  | 26.10  | .01 |                            |
| 2.08                     | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | 2.08   | .25 |                            |
| 3.87                     | 3.84   | 3.83   | 3.82   | 3.80   | 3.80   | 3.79   | 3.78   | 3.78   | 3.77   | 3.76   | 3.76   | .10 | 5                          |
| 5.86                     | 5.80   | 5.77   | 5.75   | 5.72   | 5.70   | 5.69   | 5.66   | 5.66   | 5.65   | 5.64   | 5.63   | .05 |                            |
| 14.20                    | 14.00  | 13.90  | 13.80  | 13.70  | 13.70  | 13.70  | 13.60  | 13.60  | 13.50  | 13.50  | 13.50  | .01 |                            |
| 1.89                     | 1.88   | 1.88   | 1.88   | 1.88   | 1.88   | 1.87   | 1.87   | 1.87   | 1.87   | 1.87   | 1.87   | .25 | 6                          |
| 3.24                     | 3.21   | 3.19   | 3.17   | 3.16   | 3.15   | 3.14   | 3.13   | 3.12   | 3.12   | 3.11   | 3.10   | .10 |                            |
| 4.62                     | 4.56   | 4.53   | 4.50   | 4.46   | 4.44   | 4.43   | 4.41   | 4.40   | 4.39   | 4.37   | 4.36   | .05 |                            |
| 9.72                     | 9.55   | 9.47   | 9.38   | 9.29   | 9.24   | 9.20   | 9.13   | 9.11   | 9.08   | 9.04   | 9.02   | .01 | 7                          |
| 1.76                     | 1.76   | 1.75   | 1.75   | 1.75   | 1.75   | 1.74   | 1.74   | 1.74   | 1.74   | 1.74   | 1.74   | .25 |                            |
| 2.87                     | 2.84   | 2.82   | 2.80   | 2.78   | 2.77   | 2.76   | 2.75   | 2.74   | 2.73   | 2.73   | 2.72   | .10 |                            |
| 3.94                     | 3.87   | 3.84   | 3.81   | 3.77   | 3.75   | 3.74   | 3.71   | 3.70   | 3.69   | 3.68   | 3.67   | .05 | 8                          |
| 7.56                     | 7.40   | 7.31   | 7.23   | 7.14   | 7.09   | 7.06   | 6.99   | 6.97   | 6.93   | 6.90   | 6.88   | .01 |                            |
| 1.68                     | 1.67   | 1.67   | 1.66   | 1.66   | 1.66   | 1.65   | 1.65   | 1.65   | 1.65   | 1.65   | 1.65   | .25 |                            |
| 2.63                     | 2.59   | 2.58   | 2.56   | 2.54   | 2.52   | 2.51   | 2.50   | 2.49   | 2.48   | 2.48   | 2.47   | .10 | 9                          |
| 3.51                     | 3.44   | 3.41   | 3.38   | 3.34   | 3.32   | 3.30   | 3.27   | 3.27   | 3.25   | 3.24   | 3.23   | .05 |                            |
| 6.31                     | 6.16   | 6.07   | 5.99   | 5.91   | 5.86   | 5.82   | 5.75   | 5.74   | 5.70   | 5.67   | 5.65   | .01 |                            |
| 1.62                     | 1.61   | 1.60   | 1.60   | 1.59   | 1.59   | 1.59   | 1.58   | 1.58   | 1.58   | 1.58   | 1.58   | .25 | 10                         |
| 2.46                     | 2.42   | 2.40   | 2.38   | 2.36   | 2.35   | 2.34   | 2.32   | 2.32   | 2.31   | 2.30   | 2.29   | .10 |                            |
| 3.22                     | 3.15   | 3.12   | 3.08   | 3.04   | 3.02   | 3.01   | 2.97   | 2.97   | 2.95   | 2.94   | 2.93   | .05 |                            |
| 5.52                     | 5.36   | 5.28   | 5.20   | 5.12   | 5.07   | 5.03   | 4.96   | 4.95   | 4.91   | 4.88   | 4.86   | .01 | 11                         |
| 1.57                     | 1.56   | 1.56   | 1.55   | 1.55   | 1.54   | 1.54   | 1.53   | 1.53   | 1.53   | 1.53   | 1.53   | .25 |                            |
| 2.34                     | 2.30   | 2.28   | 2.25   | 2.23   | 2.22   | 2.21   | 2.19   | 2.18   | 2.17   | 2.17   | 2.16   | .10 |                            |
| 3.01                     | 2.94   | 2.90   | 2.86   | 2.83   | 2.80   | 2.79   | 2.76   | 2.75   | 2.73   | 2.72   | 2.71   | .05 | 12                         |
| 4.96                     | 4.81   | 4.73   | 4.65   | 4.57   | 4.52   | 4.48   | 4.42   | 4.40   | 4.36   | 4.33   | 4.31   | .01 |                            |

TABLE E-3 UPPER PERCENTAGE POINTS OF THE  $F$  DISTRIBUTION (CONTINUED)

| d.f. for<br>denom-<br>inator<br>$N_2$ | d.f. for numerator $N_1$ |       |      |      |      |      |      |      |      |      |      |      |      |
|---------------------------------------|--------------------------|-------|------|------|------|------|------|------|------|------|------|------|------|
|                                       | Pr                       | 1     | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   |
| 10                                    | .25                      | 1.49  | 1.60 | 1.60 | 1.59 | 1.59 | 1.58 | 1.57 | 1.56 | 1.56 | 1.55 | 1.55 | 1.54 |
|                                       | .10                      | 3.29  | 2.92 | 2.73 | 2.61 | 2.52 | 2.46 | 2.41 | 2.38 | 2.35 | 2.32 | 2.30 | 2.28 |
|                                       | .05                      | 4.96  | 4.10 | 3.71 | 3.48 | 3.33 | 3.22 | 3.14 | 3.07 | 3.02 | 2.98 | 2.94 | 2.91 |
|                                       | .01                      | 10.00 | 7.56 | 6.55 | 5.99 | 5.64 | 5.39 | 5.20 | 5.06 | 4.94 | 4.85 | 4.77 | 4.71 |
| 11                                    | .25                      | 1.47  | 1.58 | 1.58 | 1.57 | 1.56 | 1.55 | 1.54 | 1.53 | 1.53 | 1.52 | 1.52 | 1.51 |
|                                       | .10                      | 3.23  | 2.86 | 2.66 | 2.54 | 2.45 | 2.39 | 2.34 | 2.30 | 2.27 | 2.25 | 2.23 | 2.21 |
|                                       | .05                      | 4.84  | 3.98 | 3.59 | 3.36 | 3.20 | 3.09 | 3.01 | 2.95 | 2.90 | 2.85 | 2.82 | 2.79 |
|                                       | .01                      | 9.65  | 7.21 | 6.22 | 5.67 | 5.32 | 5.07 | 4.89 | 4.74 | 4.63 | 4.54 | 4.46 | 4.40 |
| 12                                    | .25                      | 1.46  | 1.56 | 1.56 | 1.55 | 1.54 | 1.53 | 1.52 | 1.51 | 1.51 | 1.50 | 1.50 | 1.49 |
|                                       | .10                      | 3.18  | 2.81 | 2.61 | 2.48 | 2.39 | 2.33 | 2.28 | 2.24 | 2.21 | 2.19 | 2.17 | 2.15 |
|                                       | .05                      | 4.75  | 3.89 | 3.49 | 3.26 | 3.11 | 3.00 | 2.91 | 2.85 | 2.80 | 2.75 | 2.72 | 2.69 |
|                                       | .01                      | 9.33  | 6.93 | 5.95 | 5.41 | 5.06 | 4.82 | 4.64 | 4.50 | 4.39 | 4.30 | 4.22 | 4.16 |
| 13                                    | .25                      | 1.45  | 1.55 | 1.55 | 1.53 | 1.52 | 1.51 | 1.50 | 1.49 | 1.49 | 1.48 | 1.47 | 1.47 |
|                                       | .10                      | 3.14  | 2.76 | 2.56 | 2.43 | 2.35 | 2.28 | 2.23 | 2.20 | 2.16 | 2.14 | 2.12 | 2.10 |
|                                       | .05                      | 4.67  | 3.81 | 3.41 | 3.18 | 3.03 | 2.92 | 2.83 | 2.77 | 2.71 | 2.67 | 2.63 | 2.60 |
|                                       | .01                      | 9.07  | 6.70 | 5.74 | 5.21 | 4.86 | 4.62 | 4.44 | 4.30 | 4.19 | 4.10 | 4.02 | 3.96 |
| 14                                    | .25                      | 1.44  | 1.53 | 1.53 | 1.52 | 1.51 | 1.50 | 1.49 | 1.48 | 1.47 | 1.46 | 1.46 | 1.45 |
|                                       | .10                      | 3.10  | 2.73 | 2.52 | 2.39 | 2.31 | 2.24 | 2.19 | 2.15 | 2.12 | 2.10 | 2.08 | 2.05 |
|                                       | .05                      | 4.60  | 3.74 | 3.34 | 3.11 | 2.96 | 2.85 | 2.76 | 2.70 | 2.65 | 2.60 | 2.57 | 2.53 |
|                                       | .01                      | 8.86  | 6.51 | 5.56 | 5.04 | 4.69 | 4.46 | 4.28 | 4.14 | 4.03 | 3.94 | 3.86 | 3.80 |
| 15                                    | .25                      | 1.43  | 1.52 | 1.52 | 1.51 | 1.49 | 1.48 | 1.47 | 1.46 | 1.46 | 1.45 | 1.44 | 1.44 |
|                                       | .10                      | 3.07  | 2.70 | 2.49 | 2.36 | 2.27 | 2.21 | 2.16 | 2.12 | 2.09 | 2.06 | 2.04 | 2.02 |
|                                       | .05                      | 4.54  | 3.68 | 3.29 | 3.06 | 2.90 | 2.79 | 2.71 | 2.64 | 2.59 | 2.54 | 2.51 | 2.48 |
|                                       | .01                      | 8.68  | 6.36 | 5.42 | 4.89 | 4.56 | 4.32 | 4.14 | 4.00 | 3.89 | 3.80 | 3.73 | 3.67 |
| 16                                    | .25                      | 1.42  | 1.51 | 1.51 | 1.50 | 1.48 | 1.47 | 1.46 | 1.45 | 1.44 | 1.44 | 1.44 | 1.43 |
|                                       | .10                      | 3.05  | 2.67 | 2.46 | 2.33 | 2.24 | 2.18 | 2.13 | 2.09 | 2.06 | 2.03 | 2.01 | 1.99 |
|                                       | .05                      | 4.49  | 3.63 | 3.24 | 3.01 | 2.85 | 2.74 | 2.66 | 2.59 | 2.54 | 2.49 | 2.46 | 2.42 |
|                                       | .01                      | 8.53  | 6.23 | 5.29 | 4.77 | 4.44 | 4.20 | 4.03 | 3.89 | 3.78 | 3.69 | 3.62 | 3.55 |
| 17                                    | .25                      | 1.42  | 1.51 | 1.50 | 1.49 | 1.47 | 1.46 | 1.45 | 1.44 | 1.43 | 1.43 | 1.42 | 1.41 |
|                                       | .10                      | 3.03  | 2.64 | 2.44 | 2.31 | 2.22 | 2.15 | 2.10 | 2.06 | 2.03 | 2.00 | 1.98 | 1.96 |
|                                       | .05                      | 4.45  | 3.59 | 3.20 | 2.96 | 2.81 | 2.70 | 2.61 | 2.55 | 2.49 | 2.45 | 2.41 | 2.38 |
|                                       | .01                      | 8.40  | 6.11 | 5.18 | 4.67 | 4.34 | 4.10 | 3.93 | 3.79 | 3.68 | 3.59 | 3.52 | 3.46 |
| 18                                    | .25                      | 1.41  | 1.50 | 1.49 | 1.48 | 1.46 | 1.45 | 1.44 | 1.43 | 1.42 | 1.42 | 1.41 | 1.40 |
|                                       | .10                      | 3.01  | 2.62 | 2.42 | 2.29 | 2.20 | 2.13 | 2.08 | 2.04 | 2.00 | 1.98 | 1.96 | 1.93 |
|                                       | .05                      | 4.41  | 3.55 | 3.16 | 2.93 | 2.77 | 2.66 | 2.58 | 2.51 | 2.46 | 2.41 | 2.37 | 2.34 |
|                                       | .01                      | 8.29  | 6.01 | 5.09 | 4.58 | 4.25 | 4.01 | 3.84 | 3.71 | 3.60 | 3.51 | 3.43 | 3.37 |
| 19                                    | .25                      | 1.41  | 1.49 | 1.49 | 1.47 | 1.46 | 1.44 | 1.43 | 1.42 | 1.41 | 1.41 | 1.40 | 1.40 |
|                                       | .10                      | 2.99  | 2.61 | 2.40 | 2.27 | 2.18 | 2.11 | 2.06 | 2.02 | 1.98 | 1.96 | 1.94 | 1.91 |
|                                       | .05                      | 4.38  | 3.52 | 3.13 | 2.90 | 2.74 | 2.63 | 2.54 | 2.48 | 2.42 | 2.38 | 2.34 | 2.31 |
|                                       | .01                      | 8.18  | 5.93 | 5.01 | 4.50 | 4.17 | 3.94 | 3.77 | 3.63 | 3.52 | 3.43 | 3.36 | 3.30 |
| 20                                    | .25                      | 1.40  | 1.49 | 1.48 | 1.46 | 1.45 | 1.44 | 1.43 | 1.42 | 1.41 | 1.40 | 1.39 | 1.39 |
|                                       | .10                      | 2.97  | 2.59 | 2.38 | 2.25 | 2.16 | 2.09 | 2.04 | 2.00 | 1.96 | 1.94 | 1.92 | 1.89 |
|                                       | .05                      | 4.35  | 3.49 | 3.10 | 2.87 | 2.71 | 2.60 | 2.51 | 2.45 | 2.39 | 2.35 | 2.31 | 2.28 |
|                                       | .01                      | 8.10  | 5.85 | 4.94 | 4.43 | 4.10 | 3.87 | 3.70 | 3.56 | 3.46 | 3.37 | 3.29 | 3.23 |

| d.f. for numerator $N_1$ |      |      |      |      |      |      |      |      |      |      |          |     | d.f. for denominator $N_2$ |
|--------------------------|------|------|------|------|------|------|------|------|------|------|----------|-----|----------------------------|
| 15                       | 20   | 24   | 30   | 40   | 50   | 60   | 100  | 120  | 200  | 500  | $\alpha$ | Pr  |                            |
| 1.53                     | 1.52 | 1.52 | 1.51 | 1.51 | 1.50 | 1.50 | 1.49 | 1.49 | 1.49 | 1.48 | 1.48     | .25 | 10                         |
| 2.24                     | 2.20 | 2.18 | 2.16 | 2.13 | 2.12 | 2.11 | 2.09 | 2.08 | 2.07 | 2.06 | 2.06     | .10 |                            |
| 2.85                     | 2.77 | 2.74 | 2.70 | 2.66 | 2.64 | 2.62 | 2.59 | 2.58 | 2.56 | 2.55 | 2.54     | .05 |                            |
| 4.56                     | 4.41 | 4.33 | 4.25 | 4.17 | 4.12 | 4.08 | 4.01 | 4.00 | 3.96 | 3.93 | 3.91     | .01 |                            |
| 1.50                     | 1.49 | 1.49 | 1.48 | 1.47 | 1.47 | 1.47 | 1.46 | 1.46 | 1.46 | 1.45 | 1.45     | .25 | 11                         |
| 2.17                     | 2.12 | 2.10 | 2.08 | 2.05 | 2.04 | 2.03 | 2.00 | 2.00 | 1.99 | 1.98 | 1.97     | .10 |                            |
| 2.72                     | 2.65 | 2.61 | 2.57 | 2.53 | 2.51 | 2.49 | 2.46 | 2.45 | 2.43 | 2.42 | 2.40     | .05 |                            |
| 4.25                     | 4.10 | 4.02 | 3.94 | 3.86 | 3.81 | 3.78 | 3.71 | 3.69 | 3.66 | 3.62 | 3.60     | .01 |                            |
| 1.48                     | 1.47 | 1.46 | 1.45 | 1.45 | 1.44 | 1.44 | 1.43 | 1.43 | 1.43 | 1.42 | 1.42     | .25 | 12                         |
| 2.10                     | 2.06 | 2.04 | 2.01 | 1.99 | 1.97 | 1.96 | 1.94 | 1.93 | 1.92 | 1.91 | 1.90     | .10 |                            |
| 2.62                     | 2.54 | 2.51 | 2.47 | 2.43 | 2.40 | 2.38 | 2.35 | 2.34 | 2.32 | 2.31 | 2.30     | .05 |                            |
| 4.01                     | 3.86 | 3.78 | 3.70 | 3.62 | 3.57 | 3.54 | 3.47 | 3.45 | 3.41 | 3.38 | 3.36     | .01 |                            |
| 1.46                     | 1.45 | 1.44 | 1.43 | 1.42 | 1.42 | 1.42 | 1.41 | 1.41 | 1.40 | 1.40 | 1.40     | .25 | 13                         |
| 2.05                     | 2.01 | 1.98 | 1.96 | 1.93 | 1.92 | 1.90 | 1.88 | 1.88 | 1.86 | 1.85 | 1.85     | .10 |                            |
| 2.53                     | 2.46 | 2.42 | 2.38 | 2.34 | 2.31 | 2.30 | 2.26 | 2.25 | 2.23 | 2.22 | 2.21     | .05 |                            |
| 3.82                     | 3.66 | 3.59 | 3.51 | 3.43 | 3.38 | 3.34 | 3.27 | 3.25 | 3.22 | 3.19 | 3.17     | .01 |                            |
| 1.44                     | 1.43 | 1.42 | 1.41 | 1.41 | 1.40 | 1.40 | 1.39 | 1.39 | 1.39 | 1.38 | 1.38     | .25 | 14                         |
| 2.01                     | 1.96 | 1.94 | 1.91 | 1.89 | 1.87 | 1.86 | 1.83 | 1.83 | 1.82 | 1.80 | 1.80     | .10 |                            |
| 2.46                     | 2.39 | 2.35 | 2.31 | 2.27 | 2.24 | 2.22 | 2.19 | 2.18 | 2.16 | 2.14 | 2.13     | .05 |                            |
| 3.66                     | 3.51 | 3.43 | 3.35 | 3.27 | 3.22 | 3.18 | 3.11 | 3.09 | 3.06 | 3.03 | 3.00     | .01 |                            |
| 1.43                     | 1.41 | 1.41 | 1.40 | 1.39 | 1.39 | 1.38 | 1.38 | 1.37 | 1.37 | 1.36 | 1.36     | .25 | 15                         |
| 1.97                     | 1.92 | 1.90 | 1.87 | 1.85 | 1.83 | 1.82 | 1.79 | 1.79 | 1.77 | 1.76 | 1.76     | .10 |                            |
| 2.40                     | 2.33 | 2.29 | 2.25 | 2.20 | 2.18 | 2.16 | 2.12 | 2.11 | 2.10 | 2.08 | 2.07     | .05 |                            |
| 3.52                     | 3.37 | 3.29 | 3.21 | 3.13 | 3.08 | 3.05 | 2.98 | 2.96 | 2.92 | 2.89 | 2.87     | .01 |                            |
| 1.41                     | 1.40 | 1.39 | 1.38 | 1.37 | 1.37 | 1.36 | 1.36 | 1.35 | 1.35 | 1.34 | 1.34     | .25 | 16                         |
| 1.94                     | 1.89 | 1.87 | 1.84 | 1.81 | 1.79 | 1.78 | 1.76 | 1.75 | 1.74 | 1.73 | 1.72     | .10 |                            |
| 2.35                     | 2.28 | 2.24 | 2.19 | 2.15 | 2.12 | 2.11 | 2.07 | 2.06 | 2.04 | 2.02 | 2.01     | .05 |                            |
| 3.41                     | 3.26 | 3.18 | 3.10 | 3.02 | 2.97 | 2.93 | 2.86 | 2.84 | 2.81 | 2.78 | 2.75     | .01 |                            |
| 1.40                     | 1.39 | 1.38 | 1.37 | 1.36 | 1.35 | 1.35 | 1.34 | 1.34 | 1.34 | 1.33 | 1.33     | .25 | 17                         |
| 1.91                     | 1.86 | 1.84 | 1.81 | 1.78 | 1.76 | 1.75 | 1.73 | 1.72 | 1.71 | 1.69 | 1.69     | .10 |                            |
| 2.31                     | 2.23 | 2.19 | 2.15 | 2.10 | 2.08 | 2.06 | 2.02 | 2.01 | 1.99 | 1.97 | 1.96     | .05 |                            |
| 3.31                     | 3.16 | 3.08 | 3.00 | 2.92 | 2.87 | 2.83 | 2.76 | 2.75 | 2.71 | 2.68 | 2.65     | .01 |                            |
| 1.39                     | 1.38 | 1.37 | 1.36 | 1.35 | 1.34 | 1.34 | 1.33 | 1.33 | 1.32 | 1.32 | 1.32     | .25 | 18                         |
| 1.89                     | 1.84 | 1.81 | 1.78 | 1.75 | 1.74 | 1.72 | 1.70 | 1.69 | 1.68 | 1.67 | 1.66     | .10 |                            |
| 2.27                     | 2.19 | 2.15 | 2.11 | 2.06 | 2.04 | 2.02 | 1.98 | 1.97 | 1.95 | 1.93 | 1.92     | .05 |                            |
| 3.23                     | 3.08 | 3.00 | 2.92 | 2.84 | 2.78 | 2.75 | 2.68 | 2.66 | 2.62 | 2.59 | 2.57     | .01 |                            |
| 1.38                     | 1.37 | 1.36 | 1.35 | 1.34 | 1.33 | 1.33 | 1.32 | 1.32 | 1.31 | 1.31 | 1.30     | .25 | 19                         |
| 1.86                     | 1.81 | 1.79 | 1.76 | 1.73 | 1.71 | 1.70 | 1.67 | 1.67 | 1.65 | 1.64 | 1.63     | .10 |                            |
| 2.23                     | 2.16 | 2.11 | 2.07 | 2.03 | 2.00 | 1.98 | 1.94 | 1.93 | 1.91 | 1.89 | 1.88     | .05 |                            |
| 3.15                     | 3.00 | 2.92 | 2.84 | 2.76 | 2.71 | 2.67 | 2.60 | 2.58 | 2.55 | 2.51 | 2.49     | .01 |                            |
| 1.37                     | 1.36 | 1.35 | 1.34 | 1.33 | 1.33 | 1.32 | 1.31 | 1.31 | 1.30 | 1.30 | 1.29     | .25 | 20                         |
| 1.84                     | 1.79 | 1.77 | 1.74 | 1.71 | 1.69 | 1.68 | 1.65 | 1.64 | 1.63 | 1.62 | 1.61     | .10 |                            |
| 2.20                     | 2.12 | 2.08 | 2.04 | 1.99 | 1.97 | 1.95 | 1.91 | 1.90 | 1.88 | 1.86 | 1.84     | .05 |                            |
| 3.09                     | 2.94 | 2.86 | 2.78 | 2.69 | 2.64 | 2.61 | 2.54 | 2.52 | 2.48 | 2.44 | 2.42     | .01 |                            |

TABLE E-3 UPPER PERCENTAGE POINTS OF THE  $F$  DISTRIBUTION (CONTINUED)

| d.f. for<br>denom-<br>inator<br>$N_2$ | d.f. for numerator $N_1$ |      |      |      |      |      |      |      |      |      |      |      |      |
|---------------------------------------|--------------------------|------|------|------|------|------|------|------|------|------|------|------|------|
|                                       | Pr                       | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   |
| 22                                    | .25                      | 1.40 | 1.48 | 1.47 | 1.45 | 1.44 | 1.42 | 1.41 | 1.40 | 1.39 | 1.39 | 1.38 | 1.37 |
|                                       | .10                      | 2.95 | 2.56 | 2.35 | 2.22 | 2.13 | 2.06 | 2.01 | 1.97 | 1.93 | 1.90 | 1.88 | 1.86 |
|                                       | .05                      | 4.30 | 3.44 | 3.05 | 2.82 | 2.66 | 2.55 | 2.46 | 2.40 | 2.34 | 2.30 | 2.26 | 2.23 |
|                                       | .01                      | 7.95 | 5.72 | 4.82 | 4.31 | 3.99 | 3.76 | 3.59 | 3.45 | 3.35 | 3.26 | 3.18 | 3.12 |
| 24                                    | .25                      | 1.39 | 1.47 | 1.46 | 1.44 | 1.43 | 1.41 | 1.40 | 1.39 | 1.38 | 1.38 | 1.37 | 1.36 |
|                                       | .10                      | 2.93 | 2.54 | 2.33 | 2.19 | 2.10 | 2.04 | 1.98 | 1.94 | 1.91 | 1.88 | 1.85 | 1.83 |
|                                       | .05                      | 4.26 | 3.40 | 3.01 | 2.78 | 2.62 | 2.51 | 2.42 | 2.36 | 2.30 | 2.25 | 2.21 | 2.18 |
|                                       | .01                      | 7.82 | 5.61 | 4.72 | 4.22 | 3.90 | 3.67 | 3.50 | 3.36 | 3.26 | 3.17 | 3.09 | 3.03 |
| 26                                    | .25                      | 1.38 | 1.46 | 1.45 | 1.44 | 1.42 | 1.41 | 1.39 | 1.38 | 1.37 | 1.37 | 1.36 | 1.35 |
|                                       | .10                      | 2.91 | 2.52 | 2.31 | 2.17 | 2.08 | 2.01 | 1.96 | 1.92 | 1.88 | 1.86 | 1.84 | 1.81 |
|                                       | .05                      | 4.23 | 3.37 | 2.98 | 2.74 | 2.59 | 2.47 | 2.39 | 2.32 | 2.27 | 2.22 | 2.18 | 2.15 |
|                                       | .01                      | 7.72 | 5.53 | 4.64 | 4.14 | 3.82 | 3.59 | 3.42 | 3.29 | 3.18 | 3.09 | 3.02 | 2.96 |
| 28                                    | .25                      | 1.38 | 1.46 | 1.45 | 1.43 | 1.41 | 1.40 | 1.39 | 1.38 | 1.37 | 1.36 | 1.35 | 1.34 |
|                                       | .10                      | 2.89 | 2.50 | 2.29 | 2.16 | 2.06 | 2.00 | 1.94 | 1.90 | 1.87 | 1.84 | 1.81 | 1.79 |
|                                       | .05                      | 4.20 | 3.34 | 2.95 | 2.71 | 2.56 | 2.45 | 2.36 | 2.29 | 2.24 | 2.19 | 2.15 | 2.12 |
|                                       | .01                      | 7.64 | 5.45 | 4.57 | 4.07 | 3.75 | 3.53 | 3.36 | 3.23 | 3.12 | 3.03 | 2.96 | 2.90 |
| 30                                    | .25                      | 1.38 | 1.45 | 1.44 | 1.42 | 1.41 | 1.39 | 1.38 | 1.37 | 1.36 | 1.35 | 1.35 | 1.34 |
|                                       | .10                      | 2.88 | 2.49 | 2.28 | 2.14 | 2.05 | 1.98 | 1.93 | 1.88 | 1.85 | 1.82 | 1.79 | 1.77 |
|                                       | .05                      | 4.17 | 3.32 | 2.92 | 2.69 | 2.53 | 2.42 | 2.33 | 2.27 | 2.21 | 2.16 | 2.13 | 2.09 |
|                                       | .01                      | 7.56 | 5.39 | 4.51 | 4.02 | 3.70 | 3.47 | 3.30 | 3.17 | 3.07 | 2.98 | 2.91 | 2.84 |
| 40                                    | .25                      | 1.36 | 1.44 | 1.42 | 1.40 | 1.39 | 1.37 | 1.36 | 1.35 | 1.34 | 1.33 | 1.32 | 1.31 |
|                                       | .10                      | 2.84 | 2.44 | 2.23 | 2.09 | 2.00 | 1.93 | 1.87 | 1.83 | 1.79 | 1.76 | 1.73 | 1.71 |
|                                       | .05                      | 4.08 | 3.23 | 2.84 | 2.61 | 2.45 | 2.34 | 2.25 | 2.18 | 2.12 | 2.08 | 2.04 | 2.00 |
|                                       | .01                      | 7.31 | 5.18 | 4.31 | 3.83 | 3.51 | 3.29 | 3.12 | 2.99 | 2.89 | 2.80 | 2.73 | 2.66 |
| 60                                    | .25                      | 1.35 | 1.42 | 1.41 | 1.38 | 1.37 | 1.35 | 1.33 | 1.32 | 1.31 | 1.30 | 1.29 | 1.29 |
|                                       | .10                      | 2.79 | 2.39 | 2.18 | 2.04 | 1.95 | 1.87 | 1.82 | 1.77 | 1.74 | 1.71 | 1.68 | 1.66 |
|                                       | .05                      | 4.00 | 3.15 | 2.76 | 2.53 | 2.37 | 2.25 | 2.17 | 2.10 | 2.04 | 1.99 | 1.95 | 1.92 |
|                                       | .01                      | 7.08 | 4.98 | 4.13 | 3.65 | 3.34 | 3.12 | 2.95 | 2.82 | 2.72 | 2.63 | 2.56 | 2.50 |
| 120                                   | .25                      | 1.34 | 1.40 | 1.39 | 1.37 | 1.35 | 1.33 | 1.31 | 1.30 | 1.29 | 1.28 | 1.27 | 1.26 |
|                                       | .10                      | 2.75 | 2.35 | 2.13 | 1.99 | 1.90 | 1.82 | 1.77 | 1.72 | 1.68 | 1.65 | 1.62 | 1.60 |
|                                       | .05                      | 3.92 | 3.07 | 2.68 | 2.45 | 2.29 | 2.17 | 2.09 | 2.02 | 1.96 | 1.91 | 1.87 | 1.83 |
|                                       | .01                      | 6.85 | 4.79 | 3.95 | 3.48 | 3.17 | 2.96 | 2.79 | 2.66 | 2.56 | 2.47 | 2.40 | 2.34 |
| 200                                   | .25                      | 1.33 | 1.39 | 1.38 | 1.36 | 1.34 | 1.32 | 1.31 | 1.29 | 1.28 | 1.27 | 1.26 | 1.25 |
|                                       | .10                      | 2.73 | 2.33 | 2.11 | 1.97 | 1.88 | 1.80 | 1.75 | 1.70 | 1.66 | 1.63 | 1.60 | 1.57 |
|                                       | .05                      | 3.89 | 3.04 | 2.65 | 2.42 | 2.26 | 2.14 | 2.06 | 1.98 | 1.93 | 1.88 | 1.84 | 1.80 |
|                                       | .01                      | 6.76 | 4.71 | 3.88 | 3.41 | 3.11 | 2.89 | 2.73 | 2.60 | 2.50 | 2.41 | 2.34 | 2.27 |
| $\infty$                              | .25                      | 1.32 | 1.39 | 1.37 | 1.35 | 1.33 | 1.31 | 1.29 | 1.28 | 1.27 | 1.25 | 1.24 | 1.24 |
|                                       | .10                      | 2.71 | 2.30 | 2.08 | 1.94 | 1.85 | 1.77 | 1.72 | 1.67 | 1.63 | 1.60 | 1.57 | 1.55 |
|                                       | .05                      | 3.84 | 3.00 | 2.60 | 2.37 | 2.21 | 2.10 | 2.01 | 1.94 | 1.88 | 1.83 | 1.79 | 1.75 |
|                                       | .01                      | 6.63 | 4.61 | 3.78 | 3.32 | 3.02 | 2.80 | 2.64 | 2.51 | 2.41 | 2.32 | 2.25 | 2.18 |

| d.f. for numerator $N_1$ |      |      |      |      |      |      |      |      |      |      |      |     | d.f. for denominator $N_2$ |
|--------------------------|------|------|------|------|------|------|------|------|------|------|------|-----|----------------------------|
| 15                       | 20   | 24   | 30   | 40   | 50   | 60   | 100  | 120  | 200  | 500  | q    | Pr  |                            |
| 1.36                     | 1.34 | 1.33 | 1.32 | 1.31 | 1.31 | 1.30 | 1.30 | 1.30 | 1.29 | 1.29 | 1.28 | .25 | 22                         |
| 1.81                     | 1.76 | 1.73 | 1.70 | 1.67 | 1.65 | 1.64 | 1.61 | 1.60 | 1.59 | 1.58 | 1.57 | .10 |                            |
| 2.15                     | 2.07 | 2.03 | 1.98 | 1.94 | 1.91 | 1.89 | 1.85 | 1.84 | 1.82 | 1.80 | 1.78 | .05 |                            |
| 2.98                     | 2.83 | 2.75 | 2.67 | 2.58 | 2.53 | 2.50 | 2.42 | 2.40 | 2.36 | 2.33 | 2.31 | .01 |                            |
| 1.35                     | 1.33 | 1.32 | 1.31 | 1.30 | 1.29 | 1.29 | 1.28 | 1.28 | 1.27 | 1.27 | 1.26 | .25 | 24                         |
| 1.78                     | 1.73 | 1.70 | 1.67 | 1.64 | 1.62 | 1.61 | 1.58 | 1.57 | 1.56 | 1.54 | 1.53 | .10 |                            |
| 2.11                     | 2.03 | 1.98 | 1.94 | 1.89 | 1.86 | 1.84 | 1.80 | 1.79 | 1.77 | 1.75 | 1.73 | .05 |                            |
| 2.89                     | 2.74 | 2.66 | 2.58 | 2.49 | 2.44 | 2.40 | 2.33 | 2.31 | 2.27 | 2.24 | 2.21 | .01 |                            |
| 1.34                     | 1.32 | 1.31 | 1.30 | 1.29 | 1.28 | 1.28 | 1.26 | 1.26 | 1.26 | 1.25 | 1.25 | .25 | 26                         |
| 1.76                     | 1.71 | 1.68 | 1.65 | 1.61 | 1.59 | 1.58 | 1.55 | 1.54 | 1.53 | 1.51 | 1.50 | .10 |                            |
| 2.07                     | 1.99 | 1.95 | 1.90 | 1.85 | 1.82 | 1.80 | 1.76 | 1.75 | 1.73 | 1.71 | 1.69 | .05 |                            |
| 2.81                     | 2.66 | 2.58 | 2.50 | 2.42 | 2.36 | 2.33 | 2.25 | 2.23 | 2.19 | 2.16 | 2.13 | .01 |                            |
| 1.33                     | 1.31 | 1.30 | 1.29 | 1.28 | 1.27 | 1.27 | 1.26 | 1.25 | 1.25 | 1.24 | 1.24 | .25 | 28                         |
| 1.74                     | 1.69 | 1.66 | 1.63 | 1.59 | 1.57 | 1.56 | 1.53 | 1.52 | 1.50 | 1.49 | 1.48 | .10 |                            |
| 2.04                     | 1.96 | 1.91 | 1.87 | 1.82 | 1.79 | 1.77 | 1.73 | 1.71 | 1.69 | 1.67 | 1.65 | .05 |                            |
| 2.75                     | 2.60 | 2.52 | 2.44 | 2.35 | 2.30 | 2.26 | 2.19 | 2.17 | 2.13 | 2.09 | 2.06 | .01 |                            |
| 1.32                     | 1.30 | 1.29 | 1.28 | 1.27 | 1.26 | 1.26 | 1.25 | 1.24 | 1.24 | 1.23 | 1.23 | .25 | 30                         |
| 1.72                     | 1.67 | 1.64 | 1.61 | 1.57 | 1.55 | 1.54 | 1.51 | 1.50 | 1.48 | 1.47 | 1.46 | .10 |                            |
| 2.01                     | 1.93 | 1.89 | 1.84 | 1.79 | 1.76 | 1.74 | 1.70 | 1.68 | 1.66 | 1.64 | 1.62 | .05 |                            |
| 2.70                     | 2.55 | 2.47 | 2.39 | 2.30 | 2.25 | 2.21 | 2.13 | 2.11 | 2.07 | 2.03 | 2.01 | .01 |                            |
| 1.30                     | 1.28 | 1.26 | 1.25 | 1.24 | 1.23 | 1.22 | 1.21 | 1.21 | 1.20 | 1.19 | 1.19 | .25 | 40                         |
| 1.66                     | 1.61 | 1.57 | 1.54 | 1.51 | 1.48 | 1.47 | 1.43 | 1.42 | 1.41 | 1.39 | 1.38 | .10 |                            |
| 1.92                     | 1.84 | 1.79 | 1.74 | 1.69 | 1.66 | 1.64 | 1.59 | 1.58 | 1.55 | 1.53 | 1.51 | .05 |                            |
| 2.52                     | 2.37 | 2.29 | 2.20 | 2.11 | 2.06 | 2.02 | 1.94 | 1.92 | 1.87 | 1.83 | 1.80 | .01 |                            |
| 1.27                     | 1.25 | 1.24 | 1.22 | 1.21 | 1.20 | 1.19 | 1.17 | 1.17 | 1.16 | 1.15 | 1.15 | .25 | 60                         |
| 1.60                     | 1.54 | 1.51 | 1.48 | 1.44 | 1.41 | 1.40 | 1.36 | 1.35 | 1.33 | 1.31 | 1.29 | .10 |                            |
| 1.84                     | 1.75 | 1.70 | 1.65 | 1.59 | 1.56 | 1.53 | 1.48 | 1.47 | 1.44 | 1.41 | 1.39 | .05 |                            |
| 2.35                     | 2.20 | 2.12 | 2.03 | 1.94 | 1.88 | 1.84 | 1.75 | 1.73 | 1.68 | 1.63 | 1.60 | .01 |                            |
| 1.24                     | 1.22 | 1.21 | 1.19 | 1.18 | 1.17 | 1.16 | 1.14 | 1.13 | 1.12 | 1.11 | 1.10 | .25 | 120                        |
| 1.55                     | 1.48 | 1.45 | 1.41 | 1.37 | 1.34 | 1.32 | 1.27 | 1.26 | 1.24 | 1.21 | 1.19 | .10 |                            |
| 1.75                     | 1.66 | 1.61 | 1.55 | 1.50 | 1.46 | 1.43 | 1.37 | 1.35 | 1.32 | 1.28 | 1.25 | .05 |                            |
| 2.19                     | 2.03 | 1.95 | 1.86 | 1.76 | 1.70 | 1.66 | 1.56 | 1.53 | 1.48 | 1.42 | 1.38 | .01 |                            |
| 1.23                     | 1.21 | 1.20 | 1.18 | 1.16 | 1.14 | 1.12 | 1.11 | 1.10 | 1.09 | 1.08 | 1.06 | .25 | 200                        |
| 1.52                     | 1.46 | 1.42 | 1.38 | 1.34 | 1.31 | 1.28 | 1.24 | 1.22 | 1.20 | 1.17 | 1.14 | .10 |                            |
| 1.72                     | 1.62 | 1.57 | 1.52 | 1.46 | 1.41 | 1.39 | 1.32 | 1.29 | 1.26 | 1.22 | 1.19 | .05 |                            |
| 2.13                     | 1.97 | 1.89 | 1.79 | 1.69 | 1.63 | 1.58 | 1.48 | 1.44 | 1.39 | 1.33 | 1.28 | .01 |                            |
| 1.22                     | 1.19 | 1.18 | 1.16 | 1.14 | 1.13 | 1.12 | 1.09 | 1.08 | 1.07 | 1.04 | 1.00 | .25 | q                          |
| 1.49                     | 1.42 | 1.38 | 1.34 | 1.30 | 1.26 | 1.24 | 1.18 | 1.17 | 1.13 | 1.08 | 1.00 | .10 |                            |
| 1.67                     | 1.57 | 1.52 | 1.46 | 1.39 | 1.35 | 1.32 | 1.24 | 1.22 | 1.17 | 1.11 | 1.00 | .05 |                            |
| 2.04                     | 1.88 | 1.79 | 1.70 | 1.59 | 1.52 | 1.47 | 1.36 | 1.32 | 1.25 | 1.15 | 1.00 | .01 |                            |

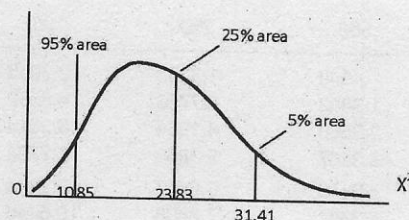
TABLE E-4 UPPER PERCENTAGE POINTS OF THE  $\chi^2$  DISTRIBUTION

## Example

$$\Pr(\chi^2 > 10.85) = 0.95$$

$$\Pr(\chi^2 > 23.83) = 0.25 \quad \text{for d.f.} = 20$$

$$\Pr(\chi^2 > 31.41) = 0.05$$



| Degrees of Freedom \ Pr | .995                     | .990                    | .975                    | .950                    | .900    |
|-------------------------|--------------------------|-------------------------|-------------------------|-------------------------|---------|
| 1                       | $392704 \times 10^{-10}$ | $157088 \times 10^{-9}$ | $982069 \times 10^{-9}$ | $393214 \times 10^{-8}$ | .0158   |
| 2                       | .0100                    | .0201                   | .0506                   | .1026                   | .2107   |
| 3                       | .0717                    | .1148                   | .2158                   | .3518                   | .5844   |
| 4                       | .2070                    | .2971                   | .4844                   | .7107                   | 1.0636  |
| 5                       | .4117                    | .5543                   | .8312                   | 1.1455                  | 1.6103  |
| 6                       | .6757                    | .8721                   | 1.2373                  | 1.6354                  | 2.2041  |
| 7                       | .9893                    | 1.2390                  | 1.6899                  | 2.1674                  | 2.8331  |
| 8                       | 1.3444                   | 1.6465                  | 2.1797                  | 2.7326                  | 3.4895  |
| 9                       | 1.7349                   | 2.0879                  | 2.7004                  | 3.3251                  | 4.1682  |
| 10                      | 2.1559                   | 2.5582                  | 3.2470                  | 3.9403                  | 4.8652  |
| 11                      | 2.6032                   | 3.0535                  | 3.8158                  | 4.5748                  | 5.5778  |
| 12                      | 3.0738                   | 3.5706                  | 4.4038                  | 5.2260                  | 6.3038  |
| 13                      | 3.5650                   | 4.1069                  | 5.0087                  | 5.8919                  | 7.0415  |
| 14                      | 4.0747                   | 4.6604                  | 5.6287                  | 6.5706                  | 7.7895  |
| 15                      | 4.6009                   | 5.2294                  | 6.2621                  | 7.2609                  | 8.5468  |
| 16                      | 5.1422                   | 5.8122                  | 6.9077                  | 7.9616                  | 9.3122  |
| 17                      | 5.6972                   | 6.4078                  | 7.5642                  | 8.6718                  | 10.0852 |
| 18                      | 6.2648                   | 7.0149                  | 8.2308                  | 9.3905                  | 10.8649 |
| 19                      | 6.8440                   | 7.6327                  | 8.9066                  | 10.1170                 | 11.6509 |
| 20                      | 7.4339                   | 8.2604                  | 9.5908                  | 10.8508                 | 12.4426 |
| 21                      | 8.0337                   | 8.8972                  | 10.2829                 | 11.5913                 | 13.2396 |
| 22                      | 8.6427                   | 9.5425                  | 10.9823                 | 12.3380                 | 14.0415 |
| 23                      | 9.2604                   | 10.1957                 | 11.6885                 | 13.0905                 | 14.8479 |
| 24                      | 9.8862                   | 10.8564                 | 12.4011                 | 13.8484                 | 15.6587 |
| 25                      | 10.5197                  | 11.5240                 | 13.1197                 | 14.6114                 | 16.4734 |
| 26                      | 11.1603                  | 12.1981                 | 13.8439                 | 15.3791                 | 17.2919 |
| 27                      | 11.8076                  | 12.8786                 | 14.5733                 | 16.1513                 | 18.1138 |
| 28                      | 12.4613                  | 13.5648                 | 15.3079                 | 16.9279                 | 18.9392 |
| 29                      | 13.1211                  | 14.2565                 | 16.0471                 | 17.7083                 | 19.7677 |
| 30                      | 13.7867                  | 14.9535                 | 16.7908                 | 18.4926                 | 20.5992 |
| 40                      | 20.7065                  | 22.1643                 | 24.4331                 | 26.5093                 | 29.0505 |
| 50                      | 27.9907                  | 29.7067                 | 32.3574                 | 34.7642                 | 37.6886 |
| 60                      | 35.5346                  | 37.4848                 | 40.4817                 | 43.1879                 | 46.4589 |
| 70                      | 43.2752                  | 45.4418                 | 48.7576                 | 51.7393                 | 55.3290 |
| 80                      | 51.1720                  | 53.5400                 | 57.1532                 | 60.3915                 | 64.2778 |
| 90                      | 59.1963                  | 61.7541                 | 65.6466                 | 69.1260                 | 73.2912 |
| 100*                    | 67.3276                  | 70.0648                 | 74.2219                 | 77.9295                 | 82.3581 |

\*For d.f. greater than 100 the expression  $\frac{2\chi^2}{2k-1} = Z$  follows the standardized normal distribution, where  $k$  represents the degrees of freedom.

| .750    | .500    | .250    | .100    | .050    | .025    | .010    | .005    |
|---------|---------|---------|---------|---------|---------|---------|---------|
| .1015   | .4549   | 1.3233  | 2.7055  | 3.8415  | 5.0239  | 6.6349  | 7.8794  |
| .5754   | 1.3863  | 2.7726  | 4.6052  | 5.9915  | 7.3778  | 9.2103  | 10.5966 |
| 1.2125  | 2.3660  | 4.1084  | 6.2514  | 7.8147  | 9.3484  | 11.3449 | 12.8381 |
| 1.9226  | 3.3567  | 5.3853  | 7.7794  | 9.4877  | 11.1433 | 13.2767 | 14.8602 |
| 2.6746  | 4.3515  | 6.6257  | 9.2364  | 11.0705 | 12.8325 | 15.0863 | 16.7496 |
| 3.4546  | 5.3481  | 7.8408  | 10.6446 | 12.5916 | 14.4494 | 16.8119 | 18.5476 |
| 4.2549  | 6.3458  | 9.0372  | 12.0170 | 14.0671 | 16.0128 | 18.4753 | 20.2777 |
| 5.0706  | 7.3441  | 10.2188 | 13.3616 | 15.5073 | 17.5346 | 20.0902 | 21.9550 |
| 5.8988  | 8.3428  | 11.3887 | 14.6837 | 16.9190 | 19.0228 | 21.6660 | 23.5893 |
| 6.7372  | 9.3418  | 12.5489 | 15.9871 | 18.3070 | 20.4831 | 23.2093 | 25.1882 |
| 7.5841  | 10.3410 | 13.7007 | 17.2750 | 19.6751 | 21.9200 | 24.7250 | 26.7569 |
| 8.4384  | 11.3403 | 14.8454 | 18.5494 | 21.0261 | 23.3367 | 26.2170 | 28.2995 |
| 9.2991  | 12.3398 | 15.9839 | 19.8119 | 22.3621 | 24.7356 | 27.6883 | 29.8194 |
| 10.1653 | 13.3393 | 17.1170 | 21.0642 | 23.6848 | 26.1190 | 29.1413 | 31.3193 |
| 11.0365 | 14.3389 | 18.2451 | 22.3072 | 24.9958 | 27.4884 | 30.5779 | 32.8013 |
| 11.9122 | 15.3385 | 19.3688 | 23.5418 | 26.2962 | 28.8454 | 31.9999 | 34.2672 |
| 12.7919 | 16.3381 | 20.4887 | 24.7690 | 27.5871 | 30.1910 | 33.4087 | 35.7185 |
| 13.6753 | 17.3379 | 21.6049 | 25.9894 | 28.8693 | 31.5264 | 34.8053 | 37.1564 |
| 14.5620 | 18.3376 | 22.7178 | 27.2036 | 30.1435 | 32.8523 | 36.1908 | 38.5822 |
| 15.4518 | 19.3374 | 23.8277 | 28.4120 | 31.4104 | 34.1696 | 37.5662 | 39.9968 |
| 16.3444 | 20.3372 | 24.9348 | 29.6151 | 32.6705 | 35.4789 | 38.9321 | 41.4010 |
| 17.2396 | 21.3370 | 26.0393 | 30.8133 | 33.9244 | 36.7807 | 40.2894 | 42.7956 |
| 18.1373 | 22.3369 | 27.1413 | 32.0069 | 35.1725 | 38.0757 | 41.6384 | 44.1813 |
| 19.0372 | 23.3367 | 28.2412 | 33.1963 | 36.4151 | 39.3641 | 42.9798 | 45.5585 |
| 19.9393 | 24.3366 | 29.3389 | 34.3816 | 37.6525 | 40.6465 | 44.3141 | 46.9278 |
| 20.8434 | 25.3364 | 30.4345 | 35.5631 | 38.8852 | 41.9232 | 45.6417 | 48.2899 |
| 21.7494 | 26.3363 | 31.5284 | 36.7412 | 40.1133 | 43.1944 | 46.9630 | 49.6449 |
| 22.6572 | 27.3363 | 32.6205 | 37.9159 | 41.3372 | 44.4607 | 48.2782 | 50.9933 |
| 23.5666 | 28.3362 | 33.7109 | 39.0875 | 42.5569 | 45.7222 | 49.5879 | 52.3356 |
| 24.4776 | 29.3360 | 34.7998 | 40.2560 | 43.7729 | 46.9792 | 50.8922 | 53.6720 |
| 33.6603 | 39.3354 | 45.6160 | 51.8050 | 55.7585 | 59.3417 | 63.6907 | 66.7659 |
| 42.9421 | 49.3349 | 56.3336 | 63.1671 | 67.5048 | 71.4202 | 76.1539 | 79.4900 |
| 52.2938 | 59.3347 | 66.9814 | 74.3970 | 79.0819 | 83.2976 | 88.3794 | 91.9517 |
| 61.6983 | 69.3344 | 77.5766 | 85.5271 | 90.5312 | 95.0231 | 100.425 | 104.215 |
| 71.1445 | 79.3343 | 88.1303 | 96.5782 | 101.879 | 106.629 | 112.329 | 116.321 |
| 80.6247 | 89.3342 | 98.6499 | 107.565 | 113.145 | 118.136 | 124.116 | 128.299 |
| 90.1332 | 99.3341 | 109.141 | 118.498 | 124.342 | 129.561 | 135.807 | 140.169 |

Source: Abridged from E. S. Pearson and H. O. Hartley, eds., *Biometrika Tables for Statisticians*, vol. 1, 3rd ed., Table 8, Cambridge University Press, New York, 1966. Reproduced by permission of the editors and trustees of *Biometrika*.

TABLE E-5a DURBIN-WATSON  $d$  STATISTIC: SIGNIFICANCE POINTS OF  $d_L$  AND  $d_U$  AT 0.05 LEVEL OF SIGNIFICANCE

| $n$ | $k' = 1$ |       | $k' = 2$ |       | $k' = 3$ |       | $k' = 4$ |       | $k' = 5$ |       | $k' = 6$ |       | $k' = 7$ |       | $k' = 8$ |       | $k' = 9$ |       | $k' = 10$ |       |
|-----|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|-----------|-------|
|     | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$     | $d_U$ |
| 6   | 0.610    | 1.400 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 7   | 0.700    | 1.356 | 0.467    | 1.896 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 8   | 0.763    | 1.332 | 0.559    | 1.777 | 0.368    | 2.287 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 9   | 0.824    | 1.320 | 0.629    | 1.699 | 0.455    | 2.128 | 0.296    | 2.588 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 10  | 0.879    | 1.320 | 0.697    | 1.641 | 0.525    | 2.016 | 0.376    | 2.414 | 0.243    | 2.822 | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 11  | 0.927    | 1.324 | 0.658    | 1.604 | 0.595    | 1.928 | 0.444    | 2.283 | 0.316    | 2.645 | 0.203    | 3.005 | —        | —     | —        | —     | —        | —     | —         | —     |
| 12  | 0.971    | 1.331 | 0.812    | 1.579 | 0.658    | 1.864 | 0.512    | 2.177 | 0.379    | 2.506 | 0.268    | 2.832 | 0.171    | 3.149 | —        | —     | —        | —     | —         | —     |
| 13  | 1.010    | 1.340 | 0.861    | 1.562 | 0.715    | 1.816 | 0.574    | 2.094 | 0.445    | 2.390 | 0.328    | 2.692 | 0.230    | 2.985 | 0.147    | 3.266 | —        | —     | —         | —     |
| 14  | 1.045    | 1.350 | 0.905    | 1.551 | 0.767    | 1.779 | 0.632    | 2.030 | 0.505    | 2.296 | 0.389    | 2.572 | 0.286    | 2.848 | 0.200    | 3.111 | 0.127    | 3.360 | —         | —     |
| 15  | 1.077    | 1.361 | 0.946    | 1.543 | 0.814    | 1.750 | 0.685    | 1.977 | 0.562    | 2.220 | 0.447    | 2.472 | 0.343    | 2.727 | 0.251    | 2.979 | 0.175    | 3.216 | 0.111     | 3.438 |
| 16  | 1.106    | 1.371 | 0.982    | 1.539 | 0.857    | 1.728 | 0.734    | 1.935 | 0.615    | 2.157 | 0.502    | 2.388 | 0.398    | 2.624 | 0.304    | 2.860 | 0.222    | 3.090 | 0.155     | 3.304 |
| 17  | 1.133    | 1.381 | 1.015    | 1.536 | 0.897    | 1.710 | 0.779    | 1.900 | 0.664    | 2.104 | 0.554    | 2.318 | 0.451    | 2.537 | 0.356    | 2.757 | 0.272    | 2.975 | 0.198     | 3.184 |
| 18  | 1.158    | 1.391 | 1.046    | 1.535 | 0.933    | 1.696 | 0.820    | 1.872 | 0.710    | 2.060 | 0.603    | 2.257 | 0.502    | 2.461 | 0.407    | 2.667 | 0.321    | 2.873 | 0.244     | 3.073 |
| 19  | 1.180    | 1.401 | 1.074    | 1.536 | 0.967    | 1.685 | 0.859    | 1.848 | 0.752    | 2.023 | 0.649    | 2.206 | 0.549    | 2.396 | 0.456    | 2.589 | 0.369    | 2.783 | 0.290     | 2.974 |
| 20  | 1.201    | 1.411 | 1.100    | 1.537 | 0.998    | 1.676 | 0.894    | 1.828 | 0.792    | 1.991 | 0.692    | 2.162 | 0.595    | 2.339 | 0.502    | 2.521 | 0.416    | 2.704 | 0.336     | 2.885 |
| 21  | 1.221    | 1.420 | 1.125    | 1.538 | 1.026    | 1.669 | 0.927    | 1.812 | 0.829    | 1.964 | 0.732    | 2.124 | 0.637    | 2.290 | 0.547    | 2.460 | 0.461    | 2.633 | 0.380     | 2.806 |
| 22  | 1.239    | 1.429 | 1.147    | 1.541 | 1.053    | 1.664 | 0.958    | 1.797 | 0.863    | 1.940 | 0.769    | 2.090 | 0.677    | 2.246 | 0.588    | 2.407 | 0.504    | 2.571 | 0.424     | 2.734 |
| 23  | 1.257    | 1.437 | 1.168    | 1.543 | 1.078    | 1.660 | 0.986    | 1.785 | 0.895    | 1.920 | 0.804    | 2.061 | 0.715    | 2.208 | 0.628    | 2.360 | 0.545    | 2.514 | 0.465     | 2.670 |
| 24  | 1.273    | 1.446 | 1.188    | 1.546 | 1.101    | 1.656 | 1.013    | 1.775 | 0.925    | 1.902 | 0.837    | 2.035 | 0.751    | 2.174 | 0.666    | 2.318 | 0.584    | 2.464 | 0.506     | 2.613 |
| 25  | 1.288    | 1.454 | 1.206    | 1.550 | 1.123    | 1.654 | 1.038    | 1.767 | 0.953    | 1.886 | 0.868    | 2.012 | 0.784    | 2.144 | 0.702    | 2.280 | 0.621    | 2.419 | 0.544     | 2.560 |
| 26  | 1.302    | 1.461 | 1.224    | 1.553 | 1.143    | 1.652 | 1.062    | 1.759 | 0.979    | 1.873 | 0.897    | 1.992 | 0.816    | 2.117 | 0.735    | 2.246 | 0.657    | 2.379 | 0.581     | 2.513 |
| 27  | 1.316    | 1.469 | 1.240    | 1.556 | 1.162    | 1.651 | 1.084    | 1.753 | 1.004    | 1.861 | 0.925    | 1.974 | 0.845    | 2.093 | 0.767    | 2.216 | 0.691    | 2.342 | 0.616     | 2.470 |
| 28  | 1.328    | 1.476 | 1.255    | 1.560 | 1.181    | 1.650 | 1.104    | 1.747 | 1.028    | 1.850 | 0.951    | 1.958 | 0.874    | 2.071 | 0.798    | 2.188 | 0.723    | 2.309 | 0.650     | 2.431 |
| 29  | 1.341    | 1.483 | 1.270    | 1.563 | 1.198    | 1.650 | 1.124    | 1.743 | 1.050    | 1.841 | 0.975    | 1.944 | 0.900    | 2.052 | 0.826    | 2.164 | 0.753    | 2.278 | 0.682     | 2.396 |
| 30  | 1.352    | 1.489 | 1.284    | 1.567 | 1.214    | 1.650 | 1.143    | 1.739 | 1.071    | 1.833 | 0.998    | 1.931 | 0.926    | 2.034 | 0.854    | 2.141 | 0.782    | 2.251 | 0.712     | 2.363 |
| 31  | 1.363    | 1.496 | 1.297    | 1.570 | 1.229    | 1.650 | 1.160    | 1.735 | 1.090    | 1.825 | 1.020    | 1.920 | 0.950    | 2.018 | 0.879    | 2.120 | 0.810    | 2.226 | 0.741     | 2.333 |
| 32  | 1.373    | 1.502 | 1.309    | 1.574 | 1.244    | 1.650 | 1.177    | 1.732 | 1.109    | 1.819 | 1.041    | 1.909 | 0.972    | 2.004 | 0.904    | 2.102 | 0.836    | 2.203 | 0.769     | 2.306 |
| 33  | 1.383    | 1.508 | 1.321    | 1.577 | 1.258    | 1.651 | 1.193    | 1.730 | 1.127    | 1.813 | 1.061    | 1.900 | 0.994    | 1.991 | 0.927    | 2.085 | 0.861    | 2.181 | 0.795     | 2.281 |
| 34  | 1.393    | 1.514 | 1.333    | 1.580 | 1.271    | 1.652 | 1.208    | 1.728 | 1.144    | 1.808 | 1.080    | 1.891 | 1.015    | 1.979 | 0.950    | 2.069 | 0.885    | 2.162 | 0.821     | 2.257 |
| 35  | 1.402    | 1.519 | 1.343    | 1.584 | 1.283    | 1.653 | 1.222    | 1.726 | 1.160    | 1.803 | 1.097    | 1.884 | 1.034    | 1.967 | 0.971    | 2.054 | 0.908    | 2.144 | 0.845     | 2.236 |
| 36  | 1.411    | 1.525 | 1.354    | 1.587 | 1.295    | 1.654 | 1.236    | 1.724 | 1.175    | 1.799 | 1.114    | 1.877 | 1.053    | 1.957 | 0.991    | 2.041 | 0.930    | 2.127 | 0.868     | 2.216 |
| 37  | 1.419    | 1.530 | 1.364    | 1.590 | 1.307    | 1.655 | 1.249    | 1.723 | 1.190    | 1.795 | 1.131    | 1.870 | 1.071    | 1.948 | 1.011    | 2.029 | 0.951    | 2.112 | 0.891     | 2.198 |
| 38  | 1.427    | 1.535 | 1.373    | 1.594 | 1.318    | 1.656 | 1.261    | 1.722 | 1.204    | 1.792 | 1.146    | 1.864 | 1.088    | 1.939 | 1.029    | 2.017 | 0.970    | 2.098 | 0.912     | 2.180 |
| 39  | 1.435    | 1.540 | 1.382    | 1.597 | 1.328    | 1.658 | 1.273    | 1.722 | 1.218    | 1.789 | 1.161    | 1.859 | 1.104    | 1.932 | 1.047    | 2.007 | 0.990    | 2.085 | 0.932     | 2.164 |
| 40  | 1.442    | 1.544 | 1.391    | 1.600 | 1.338    | 1.659 | 1.285    | 1.721 | 1.230    | 1.786 | 1.175    | 1.854 | 1.120    | 1.924 | 1.064    | 1.997 | 1.008    | 2.072 | 0.952     | 2.149 |
| 45  | 1.475    | 1.566 | 1.430    | 1.615 | 1.383    | 1.666 | 1.336    | 1.720 | 1.287    | 1.776 | 1.238    | 1.835 | 1.189    | 1.895 | 1.139    | 1.958 | 1.089    | 2.022 | 1.038     | 2.088 |
| 50  | 1.503    | 1.585 | 1.462    | 1.628 | 1.421    | 1.674 | 1.378    | 1.721 | 1.335    | 1.771 | 1.291    | 1.822 | 1.246    | 1.875 | 1.201    | 1.930 | 1.156    | 1.986 | 1.110     | 2.044 |
| 55  | 1.528    | 1.601 | 1.490    | 1.641 | 1.452    | 1.681 | 1.414    | 1.724 | 1.374    | 1.768 | 1.334    | 1.814 | 1.294    | 1.861 | 1.253    | 1.909 | 1.212    | 1.959 | 1.170     | 2.010 |
| 60  | 1.549    | 1.616 | 1.514    | 1.652 | 1.480    | 1.689 | 1.444    | 1.727 | 1.408    | 1.767 | 1.372    | 1.808 | 1.335    | 1.850 | 1.298    | 1.894 | 1.260    | 1.939 | 1.222     | 1.984 |
| 65  | 1.567    | 1.629 | 1.536    | 1.662 | 1.503    | 1.696 | 1.471    | 1.731 | 1.438    | 1.767 | 1.404    | 1.805 | 1.370    | 1.843 | 1.336    | 1.882 | 1.301    | 1.923 | 1.266     | 1.964 |
| 70  | 1.583    | 1.641 | 1.554    | 1.672 | 1.525    | 1.703 | 1.494    | 1.735 | 1.464    | 1.768 | 1.433    | 1.802 | 1.401    | 1.837 | 1.369    | 1.873 | 1.337    | 1.910 | 1.305     | 1.948 |
| 75  | 1.598    | 1.652 | 1.571    | 1.680 | 1.543    | 1.709 | 1.515    | 1.739 | 1.487    | 1.770 | 1.458    | 1.801 | 1.428    | 1.834 | 1.399    | 1.867 | 1.369    | 1.901 | 1.339     | 1.935 |
| 80  | 1.611    | 1.662 | 1.586    | 1.688 | 1.560    | 1.715 | 1.534    | 1.743 | 1.507    | 1.772 | 1.480    | 1.801 | 1.453    | 1.831 | 1.425    | 1.861 | 1.397    | 1.893 | 1.369     | 1.925 |
| 85  | 1.624    | 1.671 | 1.600    | 1.696 | 1.575    | 1.721 | 1.550    | 1.747 | 1.525    | 1.774 | 1.500    | 1.801 | 1.474    | 1.829 | 1.448    | 1.857 | 1.422    | 1.886 | 1.396     | 1.916 |
| 90  | 1.635    | 1.679 | 1.612    | 1.703 | 1.589    | 1.726 | 1.566    | 1.751 | 1.542    | 1.776 | 1.518    | 1.801 | 1.494    | 1.827 | 1.469    | 1.854 | 1.445    | 1.881 | 1.420     | 1.909 |
| 95  | 1.645    | 1.687 | 1.623    | 1.709 | 1.602    | 1.732 | 1.579    | 1.755 | 1.557    | 1.778 | 1.535    | 1.802 | 1.512    | 1.827 | 1.489    | 1.852 | 1.465    | 1.877 | 1.442     | 1.903 |
| 100 | 1.654    | 1.694 | 1.634    | 1.715 | 1.613    | 1.736 | 1.592    | 1.758 | 1.571    | 1.780 | 1.550    | 1.803 | 1.528    | 1.826 | 1.506    | 1.850 | 1.484    | 1.874 | 1.462     | 1.898 |
| 150 | 1.720    | 1.746 | 1.706    | 1.760 | 1.693    | 1.774 | 1.679    | 1.788 | 1.665    | 1.802 | 1.651    | 1.817 | 1.637    | 1.832 | 1.622    | 1.847 | 1.608    | 1.862 | 1.594     | 1.877 |
| 200 | 1.758    | 1.778 | 1.748    | 1.789 | 1.738    | 1.799 | 1.728    | 1.810 | 1.718    | 1.820 | 1.707    | 1.831 | 1.697    | 1.841 | 1.686    | 1.852 | 1.675    | 1.863 | 1.665     | 1.874 |

| n   | k' = 11        |                | k' = 12        |                | k' = 13        |                | k' = 14        |                | k' = 15        |                | k' = 16        |                | k' = 17        |                | k' = 18        |                | k' = 19        |                | k' = 20        |                |
|-----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|     | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> |
| 16  | 0.098          | 3.503          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 17  | 0.138          | 3.378          | 0.087          | 3.557          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 18  | 0.177          | 3.265          | 0.123          | 3.441          | 0.078          | 3.603          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 19  | 0.220          | 3.159          | 0.160          | 3.335          | 0.111          | 3.496          | 0.070          | 3.642          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 20  | 0.263          | 3.063          | 0.200          | 3.234          | 0.145          | 3.395          | 0.100          | 3.542          | 0.063          | 3.676          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 21  | 0.307          | 2.976          | 0.240          | 3.141          | 0.182          | 3.300          | 0.132          | 3.448          | 0.091          | 3.583          | 0.058          | 3.705          | —              | —              | —              | —              | —              | —              | —              | —              |
| 22  | 0.349          | 2.897          | 0.281          | 3.057          | 0.220          | 3.211          | 0.166          | 3.358          | 0.120          | 3.495          | 0.083          | 3.619          | 0.052          | 3.731          | —              | —              | —              | —              | —              | —              |
| 23  | 0.391          | 2.826          | 0.322          | 2.979          | 0.259          | 3.128          | 0.202          | 3.272          | 0.153          | 3.409          | 0.110          | 3.535          | 0.076          | 3.650          | 0.048          | 3.753          | —              | —              | —              | —              |
| 24  | 0.431          | 2.761          | 0.362          | 2.908          | 0.297          | 3.053          | 0.239          | 3.193          | 0.186          | 3.327          | 0.141          | 3.454          | 0.101          | 3.572          | 0.070          | 3.678          | 0.044          | 3.773          | —              | —              |
| 25  | 0.470          | 2.702          | 0.400          | 2.844          | 0.335          | 2.983          | 0.275          | 3.119          | 0.221          | 3.251          | 0.172          | 3.376          | 0.130          | 3.494          | 0.094          | 3.604          | 0.065          | 3.702          | 0.041          | 3.790          |
| 26  | 0.508          | 2.649          | 0.438          | 2.784          | 0.373          | 2.919          | 0.312          | 3.051          | 0.256          | 3.179          | 0.205          | 3.303          | 0.160          | 3.420          | 0.120          | 3.531          | 0.087          | 3.632          | 0.060          | 3.724          |
| 27  | 0.544          | 2.600          | 0.475          | 2.730          | 0.409          | 2.859          | 0.348          | 2.987          | 0.291          | 3.112          | 0.238          | 3.233          | 0.191          | 3.349          | 0.149          | 3.460          | 0.112          | 3.563          | 0.081          | 3.658          |
| 28  | 0.578          | 2.555          | 0.510          | 2.680          | 0.445          | 2.805          | 0.383          | 2.928          | 0.325          | 3.050          | 0.271          | 3.168          | 0.222          | 3.283          | 0.178          | 3.392          | 0.138          | 3.495          | 0.104          | 3.592          |
| 29  | 0.612          | 2.515          | 0.544          | 2.634          | 0.479          | 2.755          | 0.418          | 2.874          | 0.359          | 2.992          | 0.305          | 3.107          | 0.254          | 3.219          | 0.208          | 3.327          | 0.166          | 3.431          | 0.129          | 3.528          |
| 30  | 0.643          | 2.477          | 0.577          | 2.592          | 0.512          | 2.708          | 0.451          | 2.823          | 0.392          | 2.937          | 0.337          | 3.050          | 0.286          | 3.160          | 0.238          | 3.266          | 0.195          | 3.368          | 0.156          | 3.465          |
| 31  | 0.674          | 2.443          | 0.608          | 2.553          | 0.545          | 2.665          | 0.484          | 2.776          | 0.425          | 2.887          | 0.370          | 2.996          | 0.317          | 3.103          | 0.269          | 3.208          | 0.224          | 3.309          | 0.183          | 3.406          |
| 32  | 0.703          | 2.411          | 0.638          | 2.517          | 0.576          | 2.625          | 0.515          | 2.733          | 0.457          | 2.840          | 0.401          | 2.946          | 0.349          | 3.050          | 0.299          | 3.153          | 0.253          | 3.252          | 0.211          | 3.348          |
| 33  | 0.731          | 2.382          | 0.668          | 2.484          | 0.606          | 2.588          | 0.546          | 2.692          | 0.488          | 2.796          | 0.432          | 2.899          | 0.379          | 3.000          | 0.329          | 3.100          | 0.283          | 3.198          | 0.239          | 3.293          |
| 34  | 0.758          | 2.355          | 0.695          | 2.454          | 0.634          | 2.554          | 0.575          | 2.654          | 0.518          | 2.754          | 0.462          | 2.854          | 0.409          | 2.954          | 0.359          | 3.051          | 0.312          | 3.147          | 0.267          | 3.240          |
| 35  | 0.783          | 2.330          | 0.722          | 2.425          | 0.662          | 2.521          | 0.604          | 2.619          | 0.547          | 2.716          | 0.492          | 2.813          | 0.439          | 2.910          | 0.388          | 3.005          | 0.340          | 3.099          | 0.295          | 3.190          |
| 36  | 0.808          | 2.306          | 0.748          | 2.398          | 0.689          | 2.492          | 0.631          | 2.586          | 0.575          | 2.680          | 0.520          | 2.774          | 0.467          | 2.868          | 0.417          | 2.961          | 0.369          | 3.053          | 0.323          | 3.142          |
| 37  | 0.831          | 2.285          | 0.772          | 2.374          | 0.714          | 2.464          | 0.657          | 2.555          | 0.602          | 2.646          | 0.548          | 2.738          | 0.495          | 2.829          | 0.445          | 2.920          | 0.397          | 3.009          | 0.351          | 3.097          |
| 38  | 0.854          | 2.265          | 0.796          | 2.351          | 0.739          | 2.438          | 0.683          | 2.526          | 0.628          | 2.614          | 0.575          | 2.703          | 0.522          | 2.792          | 0.472          | 2.880          | 0.424          | 2.968          | 0.378          | 3.054          |
| 39  | 0.875          | 2.246          | 0.819          | 2.329          | 0.763          | 2.413          | 0.707          | 2.499          | 0.653          | 2.585          | 0.600          | 2.671          | 0.549          | 2.757          | 0.499          | 2.843          | 0.451          | 2.929          | 0.404          | 3.013          |
| 40  | 0.896          | 2.228          | 0.840          | 2.309          | 0.785          | 2.391          | 0.731          | 2.473          | 0.678          | 2.557          | 0.626          | 2.641          | 0.575          | 2.724          | 0.525          | 2.808          | 0.477          | 2.892          | 0.430          | 2.974          |
| 45  | 0.988          | 2.156          | 0.938          | 2.225          | 0.887          | 2.296          | 0.838          | 2.367          | 0.788          | 2.439          | 0.740          | 2.512          | 0.692          | 2.586          | 0.644          | 2.659          | 0.598          | 2.733          | 0.553          | 2.807          |
| 50  | 1.064          | 2.103          | 1.019          | 2.163          | 0.973          | 2.225          | 0.927          | 2.287          | 0.882          | 2.350          | 0.836          | 2.414          | 0.792          | 2.479          | 0.747          | 2.544          | 0.703          | 2.610          | 0.660          | 2.675          |
| 55  | 1.129          | 2.062          | 1.087          | 2.116          | 1.045          | 2.170          | 1.003          | 2.225          | 0.961          | 2.281          | 0.919          | 2.338          | 0.877          | 2.396          | 0.836          | 2.454          | 0.795          | 2.512          | 0.754          | 2.571          |
| 60  | 1.184          | 2.031          | 1.145          | 2.079          | 1.106          | 2.127          | 1.068          | 2.177          | 1.029          | 2.227          | 0.990          | 2.278          | 0.951          | 2.330          | 0.913          | 2.382          | 0.874          | 2.434          | 0.836          | 2.487          |
| 65  | 1.231          | 2.006          | 1.195          | 2.049          | 1.160          | 2.093          | 1.124          | 2.138          | 1.088          | 2.183          | 1.052          | 2.229          | 1.016          | 2.276          | 0.980          | 2.323          | 0.944          | 2.371          | 0.908          | 2.419          |
| 70  | 1.272          | 1.986          | 1.239          | 2.026          | 1.206          | 2.066          | 1.172          | 2.106          | 1.139          | 2.148          | 1.105          | 2.189          | 1.072          | 2.232          | 1.038          | 2.275          | 1.005          | 2.318          | 0.971          | 2.362          |
| 75  | 1.308          | 1.970          | 1.277          | 2.006          | 1.247          | 2.043          | 1.215          | 2.080          | 1.184          | 2.118          | 1.153          | 2.156          | 1.121          | 2.195          | 1.090          | 2.235          | 1.058          | 2.275          | 1.027          | 2.315          |
| 80  | 1.340          | 1.957          | 1.311          | 1.991          | 1.283          | 2.024          | 1.253          | 2.059          | 1.224          | 2.093          | 1.195          | 2.129          | 1.165          | 2.165          | 1.136          | 2.201          | 1.106          | 2.238          | 1.076          | 2.275          |
| 85  | 1.369          | 1.946          | 1.342          | 1.977          | 1.315          | 2.009          | 1.287          | 2.040          | 1.260          | 2.073          | 1.232          | 2.105          | 1.205          | 2.139          | 1.177          | 2.172          | 1.149          | 2.206          | 1.121          | 2.241          |
| 90  | 1.395          | 1.937          | 1.369          | 1.966          | 1.344          | 1.995          | 1.318          | 2.025          | 1.292          | 2.055          | 1.266          | 2.085          | 1.240          | 2.116          | 1.213          | 2.148          | 1.187          | 2.179          | 1.160          | 2.211          |
| 95  | 1.418          | 1.929          | 1.394          | 1.956          | 1.370          | 1.984          | 1.345          | 2.012          | 1.321          | 2.040          | 1.296          | 2.068          | 1.271          | 2.097          | 1.247          | 2.126          | 1.222          | 2.156          | 1.197          | 2.186          |
| 100 | 1.439          | 1.923          | 1.416          | 1.948          | 1.393          | 1.974          | 1.371          | 2.000          | 1.347          | 2.026          | 1.324          | 2.053          | 1.301          | 2.080          | 1.277          | 2.108          | 1.253          | 2.135          | 1.229          | 2.164          |
| 150 | 1.579          | 1.892          | 1.564          | 1.908          | 1.550          | 1.924          | 1.535          | 1.940          | 1.519          | 1.956          | 1.504          | 1.972          | 1.489          | 1.989          | 1.474          | 2.006          | 1.458          | 2.023          | 1.443          | 2.040          |
| 200 | 1.654          | 1.885          | 1.643          | 1.896          | 1.632          | 1.908          | 1.621          | 1.919          | 1.610          | 1.931          | 1.599          | 1.943          | 1.588          | 1.955          | 1.576          | 1.967          | 1.565          | 1.979          | 1.554          | 1.991          |

Note:  $n$  = number of observations,  $k'$  = number of explanatory variables excluding the constant term.

Source: This table is an extension of the original Durbin-Watson table and is reproduced from N.E. Savin and K.J. White, "The Durbin-Watson Test for Serial Correlation with Extreme Small Samples or Many Regressors," *Econometrica*, vol. 45, November 1977, pp. 1989-96 and as corrected by R.W. Farebrother, *Econometrica*, vol. 48, September 1980, p. 1554. Reprinted by permission of the Econometric Society.

### Example E.1.

If  $n = 40$  and  $k' = 4$ ,  $d_L = 1.285$  and  $d_U = 1.721$ . If a computed  $d$  value is less than 1.285, there is evidence of positive first-order serial correlation; if it is greater than 1.721, there is no evidence of positive first-order serial correlation; but if  $d$  lies between the lower and the upper limit, there is inconclusive evidence regarding the presence or absence of positive first-order serial correlation.

TABLE E-5b DUBIN-WATSON  $d$  STATISTIC: SIGNIFICANCE POINTS OF  $d_L$  AND  $d_U$  AT 0.01 LEVEL OF SIGNIFICANCE

| $n$ | $k' = 1$ |       | $k' = 2$ |       | $k' = 3$ |       | $k' = 4$ |       | $k' = 5$ |       | $k' = 6$ |       | $k' = 7$ |       | $k' = 8$ |       | $k' = 9$ |       | $k' = 10$ |       |
|-----|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|----------|-------|-----------|-------|
|     | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$    | $d_U$ | $d_L$     | $d_U$ |
| 6   | 0.390    | 1.142 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 7   | 0.435    | 1.036 | 0.294    | 1.676 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 8   | 0.497    | 1.003 | 0.345    | 1.489 | 0.229    | 2.102 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 9   | 0.554    | 0.998 | 0.408    | 1.389 | 0.279    | 1.875 | 0.183    | 2.433 | —        | —     | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 10  | 0.604    | 1.001 | 0.466    | 1.333 | 0.340    | 1.733 | 0.230    | 2.193 | 0.150    | 2.690 | —        | —     | —        | —     | —        | —     | —        | —     | —         | —     |
| 11  | 0.653    | 1.010 | 0.519    | 1.297 | 0.396    | 1.640 | 0.286    | 2.030 | 0.193    | 2.453 | 0.124    | 2.892 | —        | —     | —        | —     | —        | —     | —         | —     |
| 12  | 0.697    | 1.023 | 0.569    | 1.274 | 0.449    | 1.575 | 0.339    | 1.913 | 0.244    | 2.280 | 0.164    | 2.665 | 0.105    | 3.053 | —        | —     | —        | —     | —         | —     |
| 13  | 0.738    | 1.038 | 0.616    | 1.261 | 0.499    | 1.526 | 0.391    | 1.826 | 0.294    | 2.150 | 0.211    | 2.490 | 0.140    | 2.838 | 0.090    | 3.182 | —        | —     | —         | —     |
| 14  | 0.776    | 1.054 | 0.660    | 1.254 | 0.547    | 1.490 | 0.441    | 1.757 | 0.343    | 2.049 | 0.257    | 2.354 | 0.183    | 2.667 | 0.122    | 2.981 | 0.078    | 3.287 | —         | —     |
| 15  | 0.811    | 1.070 | 0.700    | 1.252 | 0.591    | 1.464 | 0.488    | 1.704 | 0.391    | 1.967 | 0.303    | 2.244 | 0.226    | 2.530 | 0.161    | 2.817 | 0.107    | 3.101 | 0.068     | 3.374 |
| 16  | 0.844    | 1.086 | 0.737    | 1.252 | 0.633    | 1.446 | 0.532    | 1.663 | 0.437    | 1.900 | 0.349    | 2.153 | 0.269    | 2.416 | 0.200    | 2.681 | 0.142    | 2.944 | 0.094     | 3.201 |
| 17  | 0.874    | 1.102 | 0.772    | 1.255 | 0.672    | 1.432 | 0.574    | 1.630 | 0.480    | 1.847 | 0.393    | 2.078 | 0.313    | 2.319 | 0.241    | 2.566 | 0.179    | 2.811 | 0.127     | 3.053 |
| 18  | 0.902    | 1.118 | 0.805    | 1.259 | 0.708    | 1.422 | 0.613    | 1.604 | 0.522    | 1.803 | 0.435    | 2.015 | 0.355    | 2.238 | 0.282    | 2.467 | 0.216    | 2.697 | 0.160     | 2.925 |
| 19  | 0.928    | 1.132 | 0.835    | 1.265 | 0.742    | 1.415 | 0.650    | 1.584 | 0.561    | 1.767 | 0.476    | 1.963 | 0.396    | 2.169 | 0.322    | 2.381 | 0.255    | 2.597 | 0.196     | 2.813 |
| 20  | 0.952    | 1.147 | 0.863    | 1.271 | 0.773    | 1.411 | 0.685    | 1.567 | 0.598    | 1.737 | 0.515    | 1.918 | 0.436    | 2.110 | 0.362    | 2.308 | 0.294    | 2.510 | 0.232     | 2.714 |
| 21  | 0.975    | 1.161 | 0.890    | 1.277 | 0.803    | 1.408 | 0.718    | 1.554 | 0.633    | 1.712 | 0.552    | 1.881 | 0.474    | 2.059 | 0.400    | 2.244 | 0.331    | 2.434 | 0.268     | 2.625 |
| 22  | 0.997    | 1.174 | 0.914    | 1.284 | 0.831    | 1.407 | 0.748    | 1.543 | 0.667    | 1.691 | 0.587    | 1.849 | 0.510    | 2.015 | 0.437    | 2.188 | 0.368    | 2.367 | 0.304     | 2.548 |
| 23  | 1.018    | 1.187 | 0.938    | 1.291 | 0.858    | 1.407 | 0.777    | 1.534 | 0.698    | 1.673 | 0.620    | 1.821 | 0.545    | 1.977 | 0.473    | 2.140 | 0.404    | 2.308 | 0.340     | 2.479 |
| 24  | 1.037    | 1.199 | 0.960    | 1.298 | 0.882    | 1.407 | 0.805    | 1.528 | 0.728    | 1.658 | 0.652    | 1.797 | 0.578    | 1.944 | 0.507    | 2.097 | 0.439    | 2.255 | 0.375     | 2.417 |
| 25  | 1.055    | 1.211 | 0.981    | 1.305 | 0.906    | 1.409 | 0.831    | 1.523 | 0.756    | 1.645 | 0.682    | 1.776 | 0.610    | 1.915 | 0.540    | 2.059 | 0.473    | 2.209 | 0.409     | 2.362 |
| 26  | 1.072    | 1.222 | 1.001    | 1.312 | 0.928    | 1.411 | 0.855    | 1.518 | 0.783    | 1.635 | 0.711    | 1.759 | 0.640    | 1.889 | 0.572    | 2.026 | 0.505    | 2.168 | 0.441     | 2.313 |
| 27  | 1.089    | 1.233 | 1.019    | 1.319 | 0.949    | 1.413 | 0.878    | 1.515 | 0.808    | 1.626 | 0.738    | 1.743 | 0.669    | 1.867 | 0.602    | 1.997 | 0.536    | 2.131 | 0.473     | 2.269 |
| 28  | 1.104    | 1.244 | 1.037    | 1.325 | 0.969    | 1.415 | 0.900    | 1.513 | 0.832    | 1.618 | 0.764    | 1.729 | 0.696    | 1.847 | 0.630    | 1.970 | 0.566    | 2.098 | 0.504     | 2.229 |
| 29  | 1.119    | 1.254 | 1.054    | 1.332 | 0.988    | 1.418 | 0.921    | 1.512 | 0.855    | 1.611 | 0.788    | 1.718 | 0.723    | 1.830 | 0.658    | 1.947 | 0.595    | 2.068 | 0.533     | 2.193 |
| 30  | 1.133    | 1.263 | 1.070    | 1.339 | 1.006    | 1.421 | 0.941    | 1.511 | 0.877    | 1.606 | 0.812    | 1.707 | 0.748    | 1.814 | 0.684    | 1.925 | 0.622    | 2.041 | 0.562     | 2.160 |
| 31  | 1.147    | 1.273 | 1.085    | 1.345 | 1.023    | 1.425 | 0.960    | 1.510 | 0.897    | 1.601 | 0.834    | 1.698 | 0.772    | 1.800 | 0.710    | 1.906 | 0.649    | 2.017 | 0.589     | 2.131 |
| 32  | 1.160    | 1.282 | 1.100    | 1.352 | 1.040    | 1.428 | 0.979    | 1.510 | 0.917    | 1.597 | 0.856    | 1.690 | 0.794    | 1.788 | 0.734    | 1.889 | 0.674    | 1.995 | 0.615     | 2.104 |
| 33  | 1.172    | 1.291 | 1.114    | 1.358 | 1.055    | 1.432 | 0.996    | 1.510 | 0.936    | 1.594 | 0.876    | 1.683 | 0.816    | 1.776 | 0.757    | 1.874 | 0.698    | 1.975 | 0.641     | 2.080 |
| 34  | 1.184    | 1.299 | 1.128    | 1.364 | 1.070    | 1.435 | 1.012    | 1.511 | 0.954    | 1.591 | 0.896    | 1.677 | 0.837    | 1.766 | 0.779    | 1.860 | 0.722    | 1.957 | 0.665     | 2.057 |
| 35  | 1.195    | 1.307 | 1.140    | 1.370 | 1.085    | 1.439 | 1.028    | 1.512 | 0.971    | 1.589 | 0.914    | 1.671 | 0.857    | 1.757 | 0.800    | 1.847 | 0.744    | 1.940 | 0.689     | 2.037 |
| 36  | 1.206    | 1.315 | 1.153    | 1.376 | 1.098    | 1.442 | 1.043    | 1.513 | 0.988    | 1.588 | 0.932    | 1.666 | 0.877    | 1.749 | 0.821    | 1.836 | 0.766    | 1.925 | 0.711     | 2.018 |
| 37  | 1.217    | 1.323 | 1.165    | 1.382 | 1.112    | 1.446 | 1.058    | 1.514 | 1.004    | 1.586 | 0.950    | 1.662 | 0.895    | 1.742 | 0.841    | 1.825 | 0.787    | 1.911 | 0.733     | 2.001 |
| 38  | 1.227    | 1.330 | 1.176    | 1.388 | 1.124    | 1.449 | 1.072    | 1.515 | 1.019    | 1.585 | 0.966    | 1.658 | 0.913    | 1.735 | 0.860    | 1.816 | 0.807    | 1.899 | 0.754     | 1.985 |
| 39  | 1.237    | 1.337 | 1.187    | 1.393 | 1.137    | 1.453 | 1.085    | 1.517 | 1.034    | 1.584 | 0.982    | 1.655 | 0.930    | 1.729 | 0.878    | 1.807 | 0.826    | 1.887 | 0.774     | 1.970 |
| 40  | 1.246    | 1.344 | 1.198    | 1.398 | 1.148    | 1.457 | 1.098    | 1.518 | 1.048    | 1.584 | 0.997    | 1.652 | 0.946    | 1.724 | 0.895    | 1.799 | 0.844    | 1.876 | 0.749     | 1.956 |
| 45  | 1.288    | 1.376 | 1.245    | 1.423 | 1.201    | 1.474 | 1.156    | 1.528 | 1.111    | 1.584 | 1.065    | 1.643 | 1.019    | 1.704 | 0.974    | 1.768 | 0.927    | 1.834 | 0.881     | 1.902 |
| 50  | 1.324    | 1.403 | 1.285    | 1.446 | 1.245    | 1.491 | 1.205    | 1.538 | 1.164    | 1.587 | 1.123    | 1.639 | 1.081    | 1.692 | 1.039    | 1.748 | 0.997    | 1.805 | 0.955     | 1.864 |
| 55  | 1.356    | 1.427 | 1.320    | 1.466 | 1.284    | 1.506 | 1.247    | 1.548 | 1.209    | 1.592 | 1.172    | 1.638 | 1.134    | 1.685 | 1.095    | 1.734 | 1.057    | 1.785 | 1.018     | 1.837 |
| 60  | 1.383    | 1.449 | 1.350    | 1.484 | 1.317    | 1.520 | 1.283    | 1.558 | 1.249    | 1.598 | 1.214    | 1.639 | 1.179    | 1.682 | 1.144    | 1.726 | 1.108    | 1.771 | 1.072     | 1.817 |
| 65  | 1.407    | 1.468 | 1.377    | 1.500 | 1.346    | 1.534 | 1.315    | 1.568 | 1.283    | 1.604 | 1.251    | 1.642 | 1.218    | 1.680 | 1.186    | 1.720 | 1.153    | 1.761 | 1.120     | 1.802 |
| 70  | 1.429    | 1.485 | 1.400    | 1.515 | 1.372    | 1.546 | 1.343    | 1.578 | 1.313    | 1.611 | 1.283    | 1.645 | 1.253    | 1.690 | 1.223    | 1.716 | 1.192    | 1.754 | 1.162     | 1.792 |
| 75  | 1.448    | 1.501 | 1.422    | 1.529 | 1.395    | 1.557 | 1.368    | 1.587 | 1.340    | 1.617 | 1.313    | 1.649 | 1.284    | 1.682 | 1.256    | 1.714 | 1.227    | 1.748 | 1.199     | 1.783 |
| 80  | 1.466    | 1.515 | 1.441    | 1.541 | 1.416    | 1.568 | 1.390    | 1.595 | 1.364    | 1.624 | 1.338    | 1.663 | 1.312    | 1.683 | 1.285    | 1.714 | 1.259    | 1.745 | 1.232     | 1.777 |
| 85  | 1.482    | 1.528 | 1.458    | 1.553 | 1.435    | 1.578 | 1.411    | 1.603 | 1.386    | 1.630 | 1.362    | 1.657 | 1.337    | 1.685 | 1.312    | 1.714 | 1.287    | 1.743 | 1.262     | 1.773 |
| 90  | 1.496    | 1.540 | 1.474    | 1.563 | 1.452    | 1.587 | 1.429    | 1.611 | 1.406    | 1.636 | 1.383    | 1.661 | 1.360    | 1.687 | 1.336    | 1.714 | 1.312    | 1.741 | 1.288     | 1.769 |
| 95  | 1.510    | 1.552 | 1.489    | 1.573 | 1.468    | 1.596 | 1.446    | 1.618 | 1.425    | 1.642 | 1.403    | 1.666 | 1.381    | 1.690 | 1.358    | 1.715 | 1.336    | 1.741 | 1.313     | 1.767 |
| 100 | 1.522    | 1.562 | 1.503    | 1.583 | 1.482    | 1.604 | 1.462    | 1.625 | 1.441    | 1.647 | 1.421    | 1.670 | 1.400    | 1.693 | 1.378    | 1.717 | 1.357    | 1.741 | 1.335     | 1.765 |
| 150 | 1.611    | 1.637 | 1.598    | 1.651 | 1.584    | 1.665 | 1.571    | 1.679 | 1.557    | 1.693 | 1.543    | 1.708 | 1.530    | 1.722 | 1.515    | 1.737 | 1.501    | 1.752 | 1.486     | 1.767 |
| 200 | 1.664    | 1.684 | 1.653    | 1.693 | 1.643    | 1.704 | 1.633    | 1.715 | 1.623    | 1.725 | 1.613    | 1.735 | 1.603    | 1.746 | 1.592    | 1.757 | 1.582    | 1.768 | 1.571     | 1.779 |

| n   | k' = 11        |                | k' = 12        |                | k' = 13        |                | k' = 14        |                | k' = 15        |                | k' = 16        |                | k' = 17        |                | k' = 18        |                | k' = 19        |                | k' = 20        |                |
|-----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|     | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> | d <sub>L</sub> | d <sub>U</sub> |
| 16  | 0.060          | 3.446          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 17  | 0.084          | 3.286          | 0.053          | 3.506          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 18  | 0.113          | 3.146          | 0.075          | 3.358          | 0.047          | 3.357          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 19  | 0.145          | 3.023          | 0.102          | 3.227          | 0.067          | 3.420          | 0.043          | 3.601          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 20  | 0.178          | 2.914          | 0.131          | 3.109          | 0.092          | 3.297          | 0.061          | 3.474          | 0.038          | 3.639          | —              | —              | —              | —              | —              | —              | —              | —              | —              | —              |
| 21  | 0.212          | 2.817          | 0.162          | 3.004          | 0.119          | 3.185          | 0.084          | 3.358          | 0.055          | 3.521          | 0.035          | 3.671          | —              | —              | —              | —              | —              | —              | —              | —              |
| 22  | 0.246          | 2.729          | 0.194          | 2.909          | 0.148          | 3.084          | 0.109          | 3.252          | 0.077          | 3.412          | 0.050          | 3.562          | 0.032          | 3.700          | —              | —              | —              | —              | —              | —              |
| 23  | 0.281          | 2.651          | 0.227          | 2.822          | 0.178          | 2.991          | 0.136          | 3.155          | 0.100          | 3.311          | 0.070          | 3.459          | 0.046          | 3.597          | 0.029          | 3.725          | —              | —              | —              | —              |
| 24  | 0.315          | 2.580          | 0.260          | 2.744          | 0.209          | 2.906          | 0.165          | 3.065          | 0.125          | 3.218          | 0.092          | 3.363          | 0.065          | 3.501          | 0.043          | 3.629          | 0.027          | 3.747          | —              | —              |
| 25  | 0.348          | 2.517          | 0.292          | 2.674          | 0.240          | 2.829          | 0.194          | 2.982          | 0.152          | 3.131          | 0.116          | 3.274          | 0.085          | 3.410          | 0.060          | 3.538          | 0.039          | 3.657          | 0.025          | 3.766          |
| 26  | 0.381          | 2.460          | 0.324          | 2.610          | 0.272          | 2.758          | 0.224          | 2.906          | 0.180          | 3.050          | 0.141          | 3.191          | 0.107          | 3.325          | 0.079          | 3.452          | 0.055          | 3.572          | 0.036          | 3.682          |
| 27  | 0.413          | 2.409          | 0.356          | 2.552          | 0.303          | 2.694          | 0.253          | 2.836          | 0.208          | 2.976          | 0.167          | 3.113          | 0.131          | 3.245          | 0.100          | 3.371          | 0.073          | 3.490          | 0.051          | 3.602          |
| 28  | 0.444          | 2.363          | 0.387          | 2.499          | 0.333          | 2.635          | 0.283          | 2.772          | 0.237          | 2.907          | 0.194          | 3.040          | 0.156          | 3.169          | 0.122          | 3.294          | 0.093          | 3.412          | 0.068          | 3.524          |
| 29  | 0.474          | 2.321          | 0.417          | 2.451          | 0.363          | 2.582          | 0.313          | 2.713          | 0.266          | 2.843          | 0.222          | 2.972          | 0.182          | 3.098          | 0.146          | 3.220          | 0.114          | 3.338          | 0.087          | 3.450          |
| 30  | 0.503          | 2.283          | 0.447          | 2.407          | 0.393          | 2.533          | 0.342          | 2.659          | 0.294          | 2.785          | 0.249          | 2.909          | 0.208          | 3.032          | 0.171          | 3.152          | 0.137          | 3.267          | 0.107          | 3.379          |
| 31  | 0.531          | 2.248          | 0.475          | 2.367          | 0.422          | 2.487          | 0.371          | 2.609          | 0.322          | 2.730          | 0.277          | 2.851          | 0.234          | 2.970          | 0.196          | 3.087          | 0.160          | 3.201          | 0.128          | 3.311          |
| 32  | 0.558          | 2.216          | 0.503          | 2.330          | 0.450          | 2.446          | 0.399          | 2.563          | 0.350          | 2.680          | 0.304          | 2.797          | 0.261          | 2.912          | 0.221          | 3.026          | 0.184          | 3.137          | 0.151          | 3.246          |
| 33  | 0.585          | 2.187          | 0.530          | 2.296          | 0.477          | 2.408          | 0.426          | 2.520          | 0.377          | 2.633          | 0.331          | 2.746          | 0.287          | 2.858          | 0.246          | 2.969          | 0.209          | 3.078          | 0.174          | 3.184          |
| 34  | 0.610          | 2.160          | 0.556          | 2.266          | 0.503          | 2.373          | 0.452          | 2.481          | 0.404          | 2.590          | 0.357          | 2.699          | 0.313          | 2.808          | 0.272          | 2.915          | 0.233          | 3.022          | 0.197          | 3.126          |
| 35  | 0.634          | 2.136          | 0.581          | 2.237          | 0.529          | 2.340          | 0.478          | 2.444          | 0.430          | 2.550          | 0.383          | 2.655          | 0.339          | 2.761          | 0.297          | 2.865          | 0.257          | 2.969          | 0.221          | 3.071          |
| 36  | 0.658          | 2.113          | 0.605          | 2.210          | 0.554          | 2.310          | 0.504          | 2.410          | 0.455          | 2.512          | 0.409          | 2.614          | 0.364          | 2.717          | 0.322          | 2.818          | 0.282          | 2.919          | 0.244          | 3.019          |
| 37  | 0.680          | 2.092          | 0.628          | 2.186          | 0.578          | 2.282          | 0.528          | 2.379          | 0.480          | 2.477          | 0.434          | 2.576          | 0.389          | 2.675          | 0.347          | 2.774          | 0.306          | 2.872          | 0.268          | 2.969          |
| 38  | 0.702          | 2.073          | 0.651          | 2.164          | 0.601          | 2.256          | 0.552          | 2.350          | 0.504          | 2.445          | 0.458          | 2.540          | 0.414          | 2.637          | 0.371          | 2.733          | 0.330          | 2.828          | 0.291          | 2.923          |
| 39  | 0.723          | 2.055          | 0.673          | 2.143          | 0.623          | 2.232          | 0.575          | 2.323          | 0.528          | 2.414          | 0.482          | 2.507          | 0.438          | 2.600          | 0.395          | 2.694          | 0.354          | 2.787          | 0.315          | 2.879          |
| 40  | 0.744          | 2.039          | 0.694          | 2.123          | 0.645          | 2.210          | 0.597          | 2.297          | 0.551          | 2.386          | 0.505          | 2.476          | 0.461          | 2.566          | 0.418          | 2.657          | 0.377          | 2.748          | 0.338          | 2.838          |
| 45  | 0.835          | 1.972          | 0.790          | 2.044          | 0.744          | 2.118          | 0.700          | 2.193          | 0.655          | 2.269          | 0.612          | 2.346          | 0.570          | 2.424          | 0.528          | 2.503          | 0.488          | 2.582          | 0.448          | 2.661          |
| 50  | 0.913          | 1.925          | 0.871          | 1.987          | 0.829          | 2.051          | 0.787          | 2.116          | 0.746          | 2.182          | 0.705          | 2.250          | 0.665          | 2.318          | 0.625          | 2.387          | 0.586          | 2.456          | 0.548          | 2.526          |
| 55  | 0.979          | 1.891          | 0.940          | 1.945          | 0.902          | 2.002          | 0.863          | 2.059          | 0.825          | 2.117          | 0.786          | 2.176          | 0.748          | 2.237          | 0.711          | 2.298          | 0.674          | 2.359          | 0.637          | 2.421          |
| 60  | 1.037          | 1.865          | 1.001          | 1.914          | 0.965          | 1.964          | 0.929          | 2.015          | 0.893          | 2.067          | 0.857          | 2.120          | 0.822          | 2.173          | 0.786          | 2.227          | 0.751          | 2.283          | 0.716          | 2.338          |
| 65  | 1.087          | 1.845          | 1.053          | 1.889          | 1.020          | 1.934          | 0.986          | 1.980          | 0.953          | 2.027          | 0.919          | 2.075          | 0.886          | 2.123          | 0.852          | 2.172          | 0.819          | 2.221          | 0.786          | 2.272          |
| 70  | 1.131          | 1.831          | 1.099          | 1.870          | 1.068          | 1.911          | 1.037          | 1.953          | 1.005          | 1.995          | 0.974          | 2.038          | 0.943          | 2.082          | 0.911          | 2.127          | 0.880          | 2.172          | 0.849          | 2.217          |
| 75  | 1.170          | 1.819          | 1.141          | 1.856          | 1.111          | 1.893          | 1.082          | 1.931          | 1.052          | 1.970          | 1.023          | 2.009          | 0.993          | 2.049          | 0.964          | 2.090          | 0.934          | 2.131          | 0.905          | 2.172          |
| 80  | 1.205          | 1.810          | 1.177          | 1.844          | 1.150          | 1.878          | 1.122          | 1.913          | 1.094          | 1.949          | 1.066          | 1.984          | 1.039          | 2.022          | 1.011          | 2.059          | 0.983          | 2.097          | 0.955          | 2.135          |
| 85  | 1.236          | 1.803          | 1.210          | 1.834          | 1.184          | 1.866          | 1.158          | 1.898          | 1.132          | 1.931          | 1.106          | 1.965          | 1.080          | 1.999          | 1.053          | 2.033          | 1.027          | 2.068          | 1.000          | 2.104          |
| 90  | 1.264          | 1.798          | 1.240          | 1.827          | 1.215          | 1.856          | 1.191          | 1.886          | 1.166          | 1.917          | 1.141          | 1.948          | 1.116          | 1.979          | 1.091          | 2.012          | 1.066          | 2.044          | 1.041          | 2.077          |
| 95  | 1.290          | 1.793          | 1.267          | 1.821          | 1.244          | 1.848          | 1.221          | 1.876          | 1.197          | 1.905          | 1.174          | 1.934          | 1.150          | 1.963          | 1.126          | 1.993          | 1.102          | 2.023          | 1.079          | 2.054          |
| 100 | 1.314          | 1.790          | 1.292          | 1.816          | 1.270          | 1.841          | 1.248          | 1.868          | 1.225          | 1.895          | 1.203          | 1.922          | 1.181          | 1.949          | 1.158          | 1.977          | 1.136          | 2.006          | 1.113          | 2.034          |
| 150 | 1.473          | 1.783          | 1.458          | 1.799          | 1.444          | 1.814          | 1.429          | 1.830          | 1.414          | 1.847          | 1.400          | 1.863          | 1.385          | 1.880          | 1.370          | 1.897          | 1.355          | 1.913          | 1.340          | 1.931          |
| 200 | 1.561          | 1.791          | 1.550          | 1.801          | 1.539          | 1.813          | 1.528          | 1.824          | 1.518          | 1.836          | 1.507          | 1.847          | 1.495          | 1.860          | 1.484          | 1.871          | 1.474          | 1.883          | 1.462          | 1.896          |

Note: n = number of observations, k' = number of explanatory variables excluding the constant term.